

# Author Response for Submission “PushAround: Collaborative Path Clearing via Physics-Informed Hybrid Search”

Author Names Omitted for Anonymous Review.

## To Associate Editor

**Q:** *The reviewers acknowledge the contribution of the paper, however, they have some concerns that need to be addressed.*

**A:** We sincerely thank the Associate Editor for the careful handling of our submission and for summarizing the reviewers’ constructive comments. In response, we have substantially revised the manuscript to clarify the methodological details, implementation choices, and experimental limitations identified by the reviewers. We also added targeted evaluations of planner–execution mismatch, obstacle-density scaling, robot-team scaling, and WCCG connectivity. The principal revisions are summarized below.

**Table 1:** New and revised experiments at a glance; page numbers refer to the revised manuscript.

| Study              | Setting                                                                                                                                      | PushAround (ours)           | Strongest baseline                    | Location                 |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|---------------------------------------|--------------------------|
| Main comparison    | 8×5 m, $M=10$ , $N=2$ ; 40 trials                                                                                                            | <b>97.5 %</b>               | SL-Push (sim) 75.0 %                  | Table II, p. 6           |
| Ablations          | Scenario 1, 40 trials                                                                                                                        | 82.5–97.5 %                 | —                                     | Table II, p. 6           |
| Obstacle density   | 8×8 m, $M=15$ –45, $N=2$ ; 10 seeds each                                                                                                     | <b>100 %</b> for all $M$    | 0/10 at $M=45$                        | Table III, Fig. 10; p. 7 |
| Physics mismatch   | 8×8 m, $M=30$ , $N=2$ ; $\pm 20\%$ and $\pm 40\%$ ; 10 seeds per condition (ours), 5 for the baseline supplements at the $\pm 40\%$ extremes | <b>97.5 %</b> at $\pm 40\%$ | DFS-WCCG 57.5 %; SL-Push (sim) 32.5 % | Table III, p. 7          |
| Robot-team scaling | 8×8 m, $M=30$ , $N=2$ –4; 10 seeds each                                                                                                      | <b>100 %</b> (30/30)        | DFS-WCCG 90–100 %; SL-Push (sim) 80 % | Table III, p. 7          |
| WCCG validation    | 300 random scenes, incl. non-convex                                                                                                          | 300/300, 0 FP/FN            | —                                     | Table I, p. 6            |

The principal revisions are:

- **Physics-model mismatch (new experiment).** The planner uses the nominal model while only the execution simulator is perturbed, one parameter family at a time, by factors covering  $\pm 20\%$  and  $\pm 40\%$ . PushAround succeeds in **156/160** perturbed trials (**97.5%**), or **166/170** (**97.6%**)

including the nominal condition, whereas under the same  $\pm 40\%$  extreme perturbations SL-Push (sim) reaches **13/40** and DFS-WCCG **23/40** (Sec. IV-A6, Table III, Fig. 1). This study is new: the first submission validated feasibility only under nominal physics, so the sensitivity of the method to an incorrect planning model had not been characterized before.

- **Density-scaling statistics (revised experiment).** The density sweep now uses **10 independent seed sequences** per density instead of five. PushAround maintains **100%** success for every  $M \in [15, 45]$  with mean planning time rising from 2.5 to 53.4 s, while the comparison methods fall to 0/10 at  $M=45$ ; dispersion and Wilson intervals are now reported (Sec. IV-A6, Table III, Fig. 7). The no-recursion ablation is reported alongside, since it isolates how much of the dense-clutter success is attributable to recursive pushing.
- **Robot-team scaling (new experiment and mechanism).** Two mode-generation refinements were added to the shared generator, which all policies use: bounded backtracking contact assignment and a temporary per-task robot release. At  $N=4$  the observed success rates change from 70% to **100%** for PushAround, from 0% to 80% for SL-Push (sim) and from 80% to 90% for DFS-WCCG. PushAround succeeds in all 30 team-size trials at 29–30 s of planning and 97–110 s of execution, versus 102–143 s and 176–178 s for DFS-WCCG (Sec. III-C2, Sec. IV-A6, Table III, Fig. 9, and Fig. 4).
- **WCCG connectivity validation (new experiment).** Sec. III-A2 derives the criterion from  $W/2$  configuration-space inflation with explicit non-convex decomposition. On 300 scenes WCCG agrees with the continuous reference in **300/300** cases with **0 FP/FN**, while the continuous and rasterized references agree in 296/300 cases; WCCG averages **1.07 ms** per query versus 1.36 ms (CC) and 243.1 ms (FR) (Table I and Fig. 6).
- **Formulation, guarantees and interfaces.** Eq. (5) is now presented as the schedule cost guiding the search; the paper states explicitly that PIHS is a feasibility-oriented heuristic search with finite sampling, bounded lookahead, a non-admissible priority, and first-feasible termination, and makes no global optimality or completeness claim. The optimized contact forces are declared feasibility variables (not hardware setpoints), and coupled object–object motion is explained as an outcome of the joint rigid-body rollout in EvalSim (Sec. II-C, II-D, III-C2, III-C3, III-D, III-E1).
- **Comparison protocol and budgets.** Sec. IV-A1 now states the shared footprints, obstacle configurations, collision/reachability checks, success criterion and simulator, and the shared per-trial budget (**800 s** timeout, **20,000** control steps, **8** parallel workers), with unsuccessful trials assigned the 800 s cap in planning-time statistics. The previous 400 s statement is corrected.

The detailed responses below provide the manuscript locations and revised text for each reviewer comment. The supplementary evaluation package has also been updated with the per-trial records and configuration files backing the reported statistics.

## To Reviewer 3

**Q:** *This paper presents PushAround, a physics-informed hybrid search framework for collaborative path clearing in environments containing multiple movable obstacles. The problem is formulated within NAMO, but specifically considers a team of mobile robots that actively rearrange obstacles to create a corridor with sufficient clearance for a larger external vehicle.*

**A:** We sincerely thank the reviewer for the positive assessment and, in particular, for recognizing the end-to-end integration, recursive pushing, dense-clutter scalability, and hardware validation. The revision addresses the two stated weaknesses and the four follow-up questions without changing the central planning framework. The detailed responses below explain these revisions point by point and provide the exact modified text highlighted in blue. The supplementary evaluation package, including the updated simulation and hardware materials, has also been updated.

**Q:** *Strengths: - Coherent end-to-end framework; The integration of WCCG, gap ranking, pushing-mode generation, and physics validation is well designed. The method jointly considers which obstacle to move and how to push it, while validating candidate actions through simulation.*

*- Recursive pushing; The planner can first clear movable blockers in the swept region and then return to the original gap-opening task. Good feature!*

*- Scalability: The method is evaluated with up to 45 movable obstacles, which is a useful demonstration of scalability in dense clutter. The reported success remains at 100% at 45 obstacles, although the experiment uses five seeds per density.*

*- Strong simulation results and real-world validation: The proposed method substantially outperforms the reported baselines in simulation and is also demonstrated with two real robots.*

**A:** We thank the reviewer for these observations. The revision keeps the end-to-end structure unchanged and strengthens the three aspects that the revision touches: the recursive-pushing mechanism is now illustrated by its own figure (Fig. 4, Page 4) and remains a distinguishing feature of the planner; the dense-clutter evaluation now uses ten independent seed sequences per density instead of five (Sec. IV-A6, Page 7), which addresses the caveat attached to the scalability result; and the two-robot hardware demonstration, including the WCCG overlays of the executed runs, is retained in Fig. 11 (Page 8) with the execution monitor now specified explicitly.

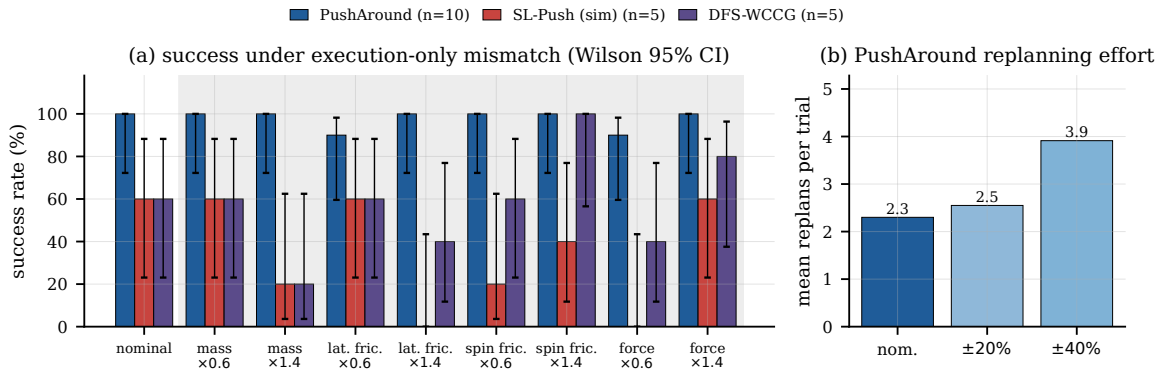
**Q:** *Physics-model robustness is insufficiently characterized: The method relies heavily on simulation-based feasibility validation, but sensitivity to uncertainty in mass, friction, force limits, contact dynamics etc. is not systematically evaluated.*

**A:** We agree that the original evaluation did not systematically isolate sensitivity to individual model parameters. The hardware experiments demonstrated closed-loop recovery but did not separate the effect of individual parameters. The revision therefore adds a dedicated planner–execution mismatch study designed to separate model mismatch from model re-identification:

- The planner always uses the nominal model; *only* the execution simulator is perturbed. The experiment therefore measures the consequence of an incorrect planning model rather than the benefit of re-identifying it.
- Obstacle mass, lateral friction, spinning friction and the robot force limit are scaled separately by  $\{0.6, 0.8, 1.2, 1.4\}$ , i.e.  $\pm 20\%$  and  $\pm 40\%$ , using the same scene seeds as the nominal condition.

Across 17 conditions ( $M=30$ ,  $N=2$ , 10 seeds each, 170 trials), PushAround succeeds in **166/170 (97.6%)**. The perturbed subset contains **156/160 successes (97.5%)**: the aggregate rate is 78/80 (97.5%) at  $\pm 20\%$  and 78/80 (97.5%) at  $\pm 40\%$ , compared with 10/10 nominal. The four failures comprise one mid-execution planning failure (lateral friction  $\times 0.6$ ) and three trials that exceeded the per-trial budget (spin friction  $\times 1.2$ , force limit  $\times 0.6$  and  $\times 1.2$ ). Mean replans increase from 2.3 (nominal) to 3.9 (averaged over the  $\pm 40\%$  conditions; up to 8.1 under force limit  $\times 0.6$ ), while mean execution time increases from 104.7 s to 159.6 s in the latter condition.

For context, SL-Push (sim) and DFS-WCCG were also evaluated under nominal planning and execution-only perturbations, using five seeds per condition. Under the eight  $\pm 40\%$  extremes, SL-Push (sim) succeeds in **13/40 trials (32.5%)** and DFS-WCCG in **23/40 trials (57.5%)**. Their mean planning times are 31.5 s and 229.1 s, respectively, compared with 18.4 s for PushAround. Because the baseline supplement uses fewer seeds and has lower nominal success (3/5 for each baseline), these results provide end-to-end context rather than a normalized estimate of each method’s degradation from its own nominal performance.



**Figure 1:** Planner-execution mismatch. (a) Success rate for PushAround ( $n=10$  per condition) and for the two baselines ( $n=5$ ) under execution-only perturbations; whiskers are Wilson 95% intervals. (b) Mean number of replans per trial for PushAround, showing that online replanning, rather than a retuned model, absorbs the mismatch.

**Changes in the revised manuscript.** Sec. I-B (Page 1) now motivates the mismatch evaluation, while Sec. IV-A6 and Table III (Page 7) define the protocol and report the perturbed and nominal aggregates:

- The revised introduction now states that model fidelity affects physics validation and that execution-model mismatch is evaluated explicitly.

Since physics validation depends on model fidelity, robustness under execution-model mismatch is also evaluated. (Sec. I-B) and Additional validation characterizes connectivity-query consistency and sensitivity to physics-model mismatch. (abstract)

- Sec. IV-A6 and Table III report 156/160 successes for the perturbed PushAround trials, distinguish the nominal condition, and give the baseline supplement results at the  $\pm 40\%$  extremes.

the planner keeps the nominal model and only the execution simulator is perturbed, one parameter at a time: PushAround succeeds in 156 of 160 perturbed trials (97.5%), the four failures being one low-friction planning failure and three budget timeouts.

**Table 2:** Complete planner–execution mismatch grid. PushAround uses 10 seeds per condition; the two baselines 5 seeds, evaluated on the nominal condition and the eight  $\pm 40\%$  extremes.  $PT/ET$  are means over successful trials,  $\bar{R}$  the mean number of replans.

| Family               | Factor | PushAround ( $n=10$ ) |      |       |           | SL-Push (sim) ( $n=5$ ) |      |       | DFS-WCCG ( $n=5$ ) |       |
|----------------------|--------|-----------------------|------|-------|-----------|-------------------------|------|-------|--------------------|-------|
|                      |        | succ.                 | $PT$ | $ET$  | $\bar{R}$ | succ.                   | $PT$ | $ET$  | succ.              | $PT$  |
| Nominal              | 1.0    | <b>10/10</b>          | 16.8 | 104.7 | 2.3       | 3/5                     | 33.6 | 310.9 | 3/5                | 156.2 |
| Mass                 | 0.6    | <b>10/10</b>          | 15.9 | 96.2  | 2.3       | 3/5                     | 30.9 | 300.4 | 3/5                | 205.6 |
| Mass                 | 0.8    | <b>10/10</b>          | 15.3 | 95.0  | 3.5       | —                       | —    | —     | —                  | —     |
| Mass                 | 1.2    | <b>10/10</b>          | 16.3 | 108.5 | 2.7       | —                       | —    | —     | —                  | —     |
| Mass                 | 1.4    | <b>10/10</b>          | 19.8 | 124.1 | 4.4       | 1/5                     | 40.2 | 319.3 | 1/5                | 136.5 |
| Lateral friction     | 0.6    | 9/10                  | 16.5 | 130.2 | 3.4       | 3/5                     | 34.4 | 315.6 | 3/5                | 222.1 |
| Lateral friction     | 0.8    | <b>10/10</b>          | 14.0 | 100.0 | 2.8       | —                       | —    | —     | —                  | —     |
| Lateral friction     | 1.2    | <b>10/10</b>          | 15.8 | 110.9 | 2.7       | —                       | —    | —     | —                  | —     |
| Lateral friction     | 1.4    | <b>10/10</b>          | 19.8 | 116.5 | 4.0       | 0/5                     | —    | —     | 2/5                | 323.2 |
| Spin friction        | 0.6    | <b>10/10</b>          | 14.1 | 91.1  | 2.7       | 1/5                     | 28.9 | 251.4 | 3/5                | 179.7 |
| Spin friction        | 0.8    | <b>10/10</b>          | 15.4 | 100.9 | 2.0       | —                       | —    | —     | —                  | —     |
| Spin friction        | 1.2    | 9/10                  | 14.9 | 98.9  | 2.2       | —                       | —    | —     | —                  | —     |
| Spin friction        | 1.4    | <b>10/10</b>          | 16.5 | 109.2 | 3.2       | 2/5                     | 27.3 | 222.6 | 5/5                | 237.0 |
| Force limit          | 0.6    | 9/10                  | 29.7 | 159.6 | 8.1       | 0/5                     | —    | —     | 2/5                | 360.3 |
| Force limit          | 0.8    | <b>10/10</b>          | 16.9 | 121.3 | 3.0       | —                       | —    | —     | —                  | —     |
| Force limit          | 1.2    | 9/10                  | 14.7 | 91.4  | 1.5       | —                       | —    | —     | —                  | —     |
| Force limit          | 1.4    | <b>10/10</b>          | 15.2 | 93.3  | 3.2       | 3/5                     | 27.6 | 301.9 | 4/5                | 168.1 |
| Aggregate $\pm 20\%$ |        | <b>78/80</b>          | 15.4 | 103.4 | 2.6       | —                       | —    | —     | —                  | —     |
| Aggregate $\pm 40\%$ |        | <b>78/80</b>          | 18.4 | 115.0 | 3.9       | 13/40                   | 31.5 | 285.2 | 23/40              | 229.1 |
| Aggregate all        |        | <b>166/170</b>        | 16.9 | 108.9 | 3.2       | 16/45                   | 31.8 | 288.9 | 26/45              | 221.0 |

**Q:** *Limited evaluation of more complex scenarios: Although the method is tested with up to 45 movable obstacles, the number of independent trials is limited.*

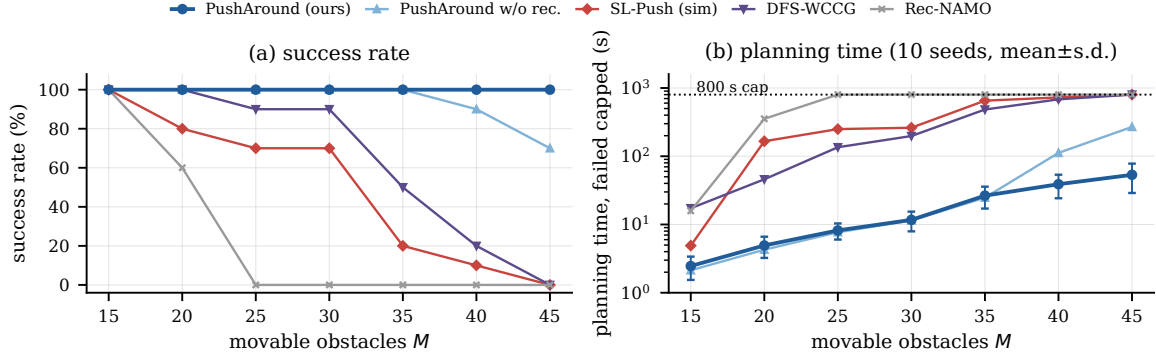
**A:** We agree, and we have both increased the number of independent runs and changed how the results are reported.

- The density sweep now uses **10 independent seed sequences** per density instead of five (7 densities  $\times$  10 seeds = 70 PushAround trials). For each seed the denser instances are generated incrementally, while different seeds are generated independently, so the success-rate estimates at neighboring densities are paired but the seeds themselves are not.
- Fig. 10 now shows the success rate and the mean planning time with one-standard-deviation whiskers; Table III lists  $PT$  and  $ET$  as mean $\pm$ s.d. Failed and pruned baseline trials are assigned the 800 s cap before averaging, and the failure composition (plan failure vs. timeout vs. timeout-pruned) is reported per density in the appendix tables.
- All headline success rates are reported with Wilson 95% intervals in Fig. 7. For PushAround at  $M=45$  the interval is  $[72.2, 100]\%$  from 10/10; for the baselines at the same density it is  $[0, 27.8]\%$  from 0/10.

**Changes in the revised manuscript.** Sec. IV-A6 and the caption of Fig. 10 (Page 7) now state the seed count, nested construction within a seed, and dispersion. Table III (Page 7) reports the corresponding mean $\pm$ s.d. values:

- Sec. IV-A6 and the caption of Fig. 10 now state the seed count, the nested construction within a seed, and the dispersion:

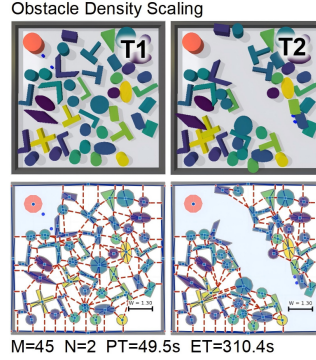
Obstacle-density scaling ( $8 \times 8$  m, 10 independent seeds per density). **(a)** Success rate; **(b)** mean planning time, with unsuccessful trials capped at the 800 s budget (dotted line); **(c)** mean



**Figure 2:** Obstacle-density scaling with 10 independent seeds per density. (a) Success rate; PushAround remains at 100% over the whole range. (b) Planning time with failed trials capped at the 800 s budget; whiskers denote one standard deviation over seeds. Rec-NAMO reaches the cap from  $M=25$  onward, and DFS-WCCG and SL-Push (sim) reach it at  $M=45$ .

**Table 3:** Density scaling, success rate (%) and capped mean planning time (s); 7 densities  $\times$  10 seeds per policy. The full table including execution time, push count and failure composition is given in Table 4.

|                             | $M=15$ | 20    | 25    | 30    | 35    | 40    | 45         |
|-----------------------------|--------|-------|-------|-------|-------|-------|------------|
| PushAround (succ.)          | 100    | 100   | 100   | 100   | 100   | 100   | <b>100</b> |
| PushAround (PT)             | 2.5    | 4.9   | 8.2   | 11.7  | 26.5  | 38.9  | 53.4       |
| PushAround w/o rec. (succ.) | 100    | 100   | 100   | 100   | 100   | 90    | 70         |
| PushAround w/o rec. (PT)    | 2.1    | 4.3   | 7.7   | 11.4  | 24.7  | 112.0 | 268.9      |
| SL-Push (sim) (succ.)       | 100    | 80    | 70    | 70    | 20    | 10    | 0          |
| SL-Push (sim) (PT)          | 4.9    | 165.9 | 249.9 | 261.5 | 652.5 | 727.0 | 800.0      |
| DFS-WCCG (succ.)            | 100    | 100   | 90    | 90    | 50    | 20    | 0          |
| DFS-WCCG (PT)               | 17.3   | 45.9  | 135.5 | 197.7 | 483.8 | 681.7 | 800.0      |
| Rec-NAMO (succ.)            | 100    | 60    | 0     | 0     | 0     | 0     | 0          |
| Rec-NAMO (PT)               | 15.8   | 353.5 | 800.0 | 800.0 | 800.0 | 800.0 | 800.0      |



**Figure 3:** Representative dense-clutter scenes of the density sweep ( $M=45$ , two robots): snapshots during clearing above and the corresponding WCCG overlays with the start face in blue below.

execution time of the successful trials (Rec-NAMO has none beyond  $M=20$ ). Error bars denote one standard deviation across seeds.

- Table III reports  $PT/ET$  as  $\text{mean} \pm \text{s.d.}$  for the representative densities  $M \in \{15, 30, 45\}$ .

**Table 4:** Complete obstacle-density grid (10 seeds per cell). Top: success rate (%). Bottom: mean planning time with unsuccessful trials capped at 800 s. PushAround additionally reports execution time and push count.

| Success (%)               | $M=15$          | 20                | 25                | 30                | 35                | 40                | 45                |
|---------------------------|-----------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| PushAround                | 100             | 100               | 100               | 100               | 100               | 100               | 100               |
| PushAround w/o rec.       | 100             | 100               | 100               | 100               | 100               | 90                | 70                |
| SL-Push (sim)             | 100             | 80                | 70                | 70                | 20                | 10                | 0                 |
| DFS-WCCG                  | 100             | 100               | 90                | 90                | 50                | 20                | 0                 |
| Rec-NAMO                  | 100             | 60                | 0                 | 0                 | 0                 | 0                 | 0                 |
| Planning time (s)         | $M=15$          | 20                | 25                | 30                | 35                | 40                | 45                |
| PushAround                | $2.5 \pm 0.9$   | $4.9 \pm 1.7$     | $8.2 \pm 2.2$     | $11.7 \pm 3.7$    | $26.5 \pm 9.3$    | $38.9 \pm 14.7$   | $53.4 \pm 24.5$   |
| PushAround w/o rec.       | $2.1 \pm 0.8$   | $4.3 \pm 1.3$     | $7.7 \pm 2.1$     | $11.4 \pm 3.0$    | $24.7 \pm 8.2$    | $112.0 \pm 242.0$ | $268.9 \pm 366.6$ |
| SL-Push (sim)             | $4.9 \pm 4.8$   | $165.9 \pm 334.2$ | $249.9 \pm 379.7$ | $261.5 \pm 371.9$ | $652.5 \pm 311.5$ | $727.0 \pm 230.9$ | 800.0             |
| DFS-WCCG                  | $17.3 \pm 9.1$  | $45.9 \pm 47.5$   | $135.5 \pm 241.1$ | $197.7 \pm 248.4$ | $483.8 \pm 339.6$ | $681.7 \pm 253.5$ | 800.0             |
| Rec-NAMO                  | $15.8 \pm 12.4$ | $353.5 \pm 385.5$ | 800.0             | 800.0             | 800.0             | 800.0             | 800.0             |
| PushAround: <i>ET</i> (s) | 37.1            | 56.9              | 77.3              | 100.9             | 165.4             | 213.2             | 272.7             |
| PushAround: pushes        | 3.5             | 5.8               | 7.9               | 10.5              | 16.6              | 20.7              | 26.1              |

**Q:** *The experiments primarily use two robots. What happens when the number of robots increases with the complexity of the task? How does the search and the contact/force formulation scale to larger robot teams?*

**A:** This comment revealed a bottleneck in the shared mode-generation module. We added two local refinements and re-ran PushAround and both baselines with the same updated generator; the resulting comparison is reported below.

For complexity, with  $|\mathcal{A}|$  reachable contact candidates per robot, an exhaustive joint assignment would require  $|\mathcal{A}|^N$  tuples. The greedy stage instead tests at most  $N|\mathcal{A}|$  single-contact substitutions; when it fails, the bounded-backtracking search expands at most  $B=2000$  partial assignments — a fixed budget that does not grow with the team size — and the subset fallback repeats that search for at most one excluded robot. Each resulting candidate is checked by a force LP with a  $2N$ -dimensional decision vector. The number of contact assignments examined is bounded by  $N|\mathcal{A}|+2B$ , rather than growing as  $|\mathcal{A}|^N$ ; each examined assignment requires an LP over  $2N$  force variables, so team size affects both the number and the dimension of the LPs, while the implementation avoids exhaustive enumeration of joint contact tuples. Sec. III-E2 states this distinction and the budget used.

The preliminary  $N=4$  runs failed in 3 of 10 seeds (2 planning failures and 1 budget timeout). The diagnosis was that, for some push tasks *no* full-team contact assignment exists — small pushable faces, or the pairwise pusher-clearance constraint rejecting every remaining candidate of one robot — and the legacy greedy replacement was additionally sensitive to robot ordering. We added two mode-generation-local refinements that leave global planning, task sampling and the search tree unchanged: (i) *bounded backtracking* contact assignment and (ii) a *temporary per-task robot release* (up to one robot is excluded for that single push and holds its pose, a penalty is charged during mode ranking so full-team modes are preferred, and the full team is re-available for every later task).

Both refinements are in the shared mode generator, so the baseline cells were re-run with the same code. Across their 40 trial logs, instrumentation records 20,890 failed greedy replacements, 10,904 bounded-backtracking assignments (of which 739 rescue a failed greedy replacement), and 6,017 per-task robot releases. These counts confirm that the added paths are exercised by the baselines as well as by PushAround:

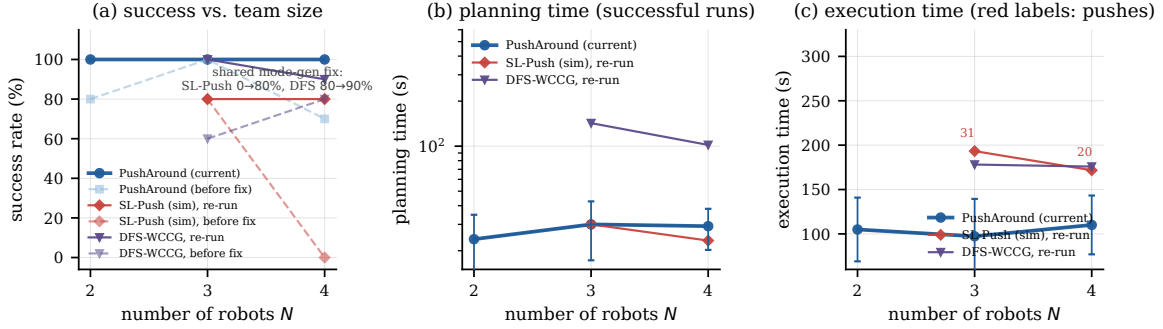
After the refinement, all three methods solve 80–100% of the team-size trials, and the success-rate intervals at  $N=3, 4$  overlap (at  $N=4$ : PushAround  $[0.72, 1.00]$ , DFS-WCCG  $[0.60, 0.98]$ , and SL-Push (sim)  $[0.49, 0.94]$ ). We therefore make no monotone team-size claim. The observed separation

**Table 5:** Robot-team scaling at  $M=30$  (nominal physics) after the shared mode-generation fix. Every cell uses 10 seeds and the same shared mode-generation implementation and evaluation protocol;  $PT/ET$  are means over successful trials. For reference, the pre-fix history is given in the notes.

| Policy        | $N$ | Trials | Succ.        | $PT$ (s)          | $ET$ (s)          | pushes | sims |
|---------------|-----|--------|--------------|-------------------|-------------------|--------|------|
| PushAround    | 2   | 10     | <b>10/10</b> | $23.9 \pm 10.9$   | $105.0 \pm 35.9$  | 11.7   | 81   |
|               | 3   | 10     | <b>10/10</b> | $30.0 \pm 12.7$   | $97.3 \pm 42.1$   | 11.1   | 66   |
|               | 4   | 10     | <b>10/10</b> | $29.2 \pm 8.9$    | $110.1 \pm 33.1$  | 10.0   | 59   |
| SL-Push (sim) | 3   | 10     | 8/10         | $30.0 \pm 46.3$   | $193.3 \pm 111.3$ | 31.1   | 62   |
| SL-Push (sim) | 4   | 10     | 8/10         | $23.4 \pm 18.2$   | $171.7 \pm 61.2$  | 20.5   | 41   |
| DFS-WCCG      | 3   | 10     | <b>10/10</b> | $142.6 \pm 183.1$ | $178.1 \pm 72.1$  | 13.9   | 762  |
| DFS-WCCG      | 4   | 10     | 9/10         | $101.9 \pm 103.9$ | $175.9 \pm 113.6$ | 14.3   | 498  |

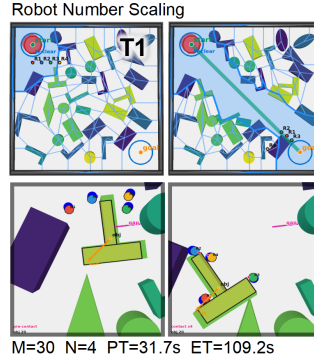
Failure composition: SL-Push (sim)  $N=3$  one plan failure and one timeout,  $N=4$  two timeouts; DFS-WCCG  $N=4$  one timeout; PushAround none.

Pre-fix history (superseded; baselines  $n=5$ ): at  $N=3$  PushAround 10/10, SL-Push (sim) 4/5, DFS-WCCG 3/5; at  $N=4$  PushAround 7/10, SL-Push (sim) 0/5 (all plan failures), DFS-WCCG 4/5.



**Figure 4:** Robot-team scaling after the shared mode-generation refinement, with the pre-fix runs shown as dashed history. At  $N=4$ , the observed success rate is 100% for PushAround, 80% for SL-Push (sim), and 90% for DFS-WCCG. (b) Planning time and (c) execution time of successful runs; the annotations in (c) are the push counts of SL-Push (sim).

is descriptive: PushAround is the only method with 30/30 successes over  $N=2, 3, 4$ ; at  $N=3, 4$  it uses 29–30 s of planning and 97–110 s of execution, compared with 102–143 s and 176–178 s for DFS-WCCG, while SL-Push (sim) requires 172–193 s of execution and 20–31 pushes.



**Figure 5:** The four-robot study of the same evaluation ( $M=30$ ,  $N=4$ ): global initial and final states above, and two magnified views of a single pushing task (before contact and during the push) below.

**Changes in the revised manuscript.** Sec. III-C2 (Page 4) now describes the shared mode-generation

refinements and their preference for full-team assignments. Sec. IV-A6 and Table III (Page 7) report the post-fix  $N \in \{2, 3, 4\}$  comparison for all policies, while Sec. III-E2 (Page 5) explains the bounded contact-assignment search:

- The revised text explains that bounded backtracking reduces the robot-order sensitivity of greedy replacement, and that a temporary per-task robot release is penalized so full-team modes remain preferred.

A bounded backtracking variant removes the robot-order sensitivity of this replacement. If no full-team assignment exists for a task, one robot is released for that single push and the full team is available again afterwards.

- The robot-team group of Table III reports success, planning time, and execution time for PushAround and both baselines at  $N=3$  and  $N=4$ , using 10 seeds per cell.
- Sec. III-E2 contrasts the  $N|\mathcal{A}|$  greedy stage with the  $|\mathcal{A}|^N$  exhaustive assignment space and notes the finite backtracking budget.

the greedy stage evaluates at most  $N|\mathcal{A}|$  single-contact substitutions instead of the exhaustive  $|\mathcal{A}|^N$  assignment, and when it fails, a bounded-backtracking search expands at most  $B=2000$  partial assignments, a fixed budget independent of the team size. The per-task cost is therefore at most  $(N|\mathcal{A}|+2B)$  LP feasibility checks, rather than growing as  $|\mathcal{A}|^N$ .

**Table 6:** Complete robot-team-scaling grid ( $M=30$ , nominal physics), all cells on the post-fix code with 10 seeds. The pre-fix rows are kept only as history.

| Policy                                                          | $N$ | Trials | Succ. | $PT$ (s)          | $ET$ (s)          | pushes | sims |
|-----------------------------------------------------------------|-----|--------|-------|-------------------|-------------------|--------|------|
| PushAround (current)                                            | 2   | 10     | 10/10 | $23.9 \pm 10.9$   | $105.0 \pm 35.9$  | 11.7   | 81   |
| PushAround (current)                                            | 3   | 10     | 10/10 | $30.0 \pm 12.7$   | $97.3 \pm 42.1$   | 11.1   | 66   |
| PushAround (current)                                            | 4   | 10     | 10/10 | $29.2 \pm 8.9$    | $110.1 \pm 33.1$  | 10.0   | 59   |
| SL-Push (sim)                                                   | 3   | 10     | 8/10  | $30.0 \pm 46.3$   | $193.3 \pm 111.3$ | 31.1   | 62   |
| SL-Push (sim)                                                   | 4   | 10     | 8/10  | $23.4 \pm 18.2$   | $171.7 \pm 61.2$  | 20.5   | 41   |
| DFS-WCCG                                                        | 3   | 10     | 10/10 | $142.6 \pm 183.1$ | $178.1 \pm 72.1$  | 13.9   | 762  |
| DFS-WCCG                                                        | 4   | 10     | 9/10  | $101.9 \pm 103.9$ | $175.9 \pm 113.6$ | 14.3   | 498  |
| <i>Pre-fix history (superseded; baselines <math>n=5</math>)</i> |     |        |       |                   |                   |        |      |
| PushAround                                                      | 2   | 10     | 8/10  | 19.5              | 125.6             | —      | —    |
| PushAround                                                      | 3   | 10     | 10/10 | 21.8              | 100.6             | —      | —    |
| PushAround                                                      | 4   | 10     | 7/10  | 23.3              | 121.3             | —      | —    |
| SL-Push (sim)                                                   | 3   | 5      | 4/5   | 61.7              | 196.7             | —      | —    |
| SL-Push (sim)                                                   | 4   | 5      | 0/5   | —                 | —                 | —      | —    |
| DFS-WCCG                                                        | 3   | 5      | 3/5   | 164.6             | 199.8             | —      | —    |
| DFS-WCCG                                                        | 4   | 5      | 4/5   | 131.0             | 190.9             | —      | —    |

**Q:** What guarantees, if any, does the hybrid search provide regarding solution quality?

**A:** We appreciate this question. No global optimality or completeness guarantee is claimed, and the revision now states this explicitly in the formulation, node-selection, and termination discussions.

PIHS samples a finite set of pushing directions and contact modes, uses bounded gap lookahead, ranks nodes with a heuristic best-first priority whose remaining-cost term is *not* required to be admissible, and terminates at the first physics-validated  $W$ -clear solution. The returned schedule is therefore a feasible schedule under the evaluation model, not a globally minimal one. Eq. (5) is consequently presented as the clearance cost that ranks candidate schedules; it does not turn the implemented search into a solver of the underlying continuous hybrid optimization problem.

Completeness is also not guaranteed: a feasible scene may require an unsampled pushing direction, contact assignment, or deeper recursion, or may exceed the finite computation budget. The node-expansion counts and ablations characterize empirical search efficiency and solution-finding reliability; they are not guarantees of solution quality.

**Changes in the revised manuscript.** Sec. II-D (Page 2), the node-selection discussion (Page 5), and the termination discussion (Page 5) now describe Eq. (5) as a ranking cost and explicitly state the limitations introduced by finite sampling, bounded lookahead, a non-admissible heuristic, and first-feasible termination:

- The formulation now describes Eq. (5) as ranking feasible schedules and limits the claim to simulator-validated feasibility.

Equation (5) ranks feasible schedules; PIHS uses finite sampling and does not guarantee global optimality of the continuous hybrid objective.

- The search and termination text state that finite sampling and a finite computation budget do not guarantee completeness.

The search operates on a finite sampled candidate set and therefore does not guarantee global optimality or completeness of the underlying continuous hybrid planning problem. (Sec. III-D) and the schedule  $\pi$  stored in  $\nu$  is a feasible solution satisfying the terminal clearance condition and the validated pushing transitions, without implying that  $\mathcal{J}_{\text{clr}}(\pi)$  is globally minimal. The objective covers the clearing phase only, since the external vehicle traverses the corridor after clearing completes. (termination)

- The node-selection text identifies the remaining-cost term as a non-admissible heuristic and makes clear that the priority does not provide an A\* optimality guarantee.

The remaining-cost term is a heuristic and is not required to be admissible. The score prioritizes nodes by realized cost and promising future actions, but it does not provide an A\* optimality guarantee.

**Q:** The objective includes the total duration  $T$  and the control-effort cost  $J$ . Is the time needed for the final vehicle to actually traverse the cleared corridor included in this objective, or is the objective only concerned with clearing the obstacles? Similarly, does  $J$  only capture the robots' effort in moving the obstacles?

**A:** Yes. The objective covers the robotic clearance operation only, while the vehicle traversal appears only through the terminal clearance condition.

- $T \triangleq \sum_{k=1}^K \Delta t_k$  is the duration of robotic clearance only; the external vehicle is not moving concurrently and traverses the corridor after clearing is complete.
- $J(\cdot)$  is the simulation-estimated manipulation effort for moving the obstacles. Vehicle traversal time is not part of the objective and is represented only by the terminal  $W$ -clearance condition.

**Changes in the revised manuscript:** Sec. II-D (Page 2) now defines  $T$  as the robotic clearance duration, defines  $J(\cdot)$  as the simulation-estimated manipulation effort, and states that the external vehicle traverses only after clearing and is represented through the terminal  $W$ -clearance condition.

**Q:** *The paper demonstrates online replanning when the real execution differs from the simulation, How can simple pushing robots evaluate this discrepancy?*

**A:** We agree that the previous wording (“the realized motion deviates from simulation”) was too vague and could be read as requiring online parameter identification or force sensing. That is not the implementation, and the revision replaces the phrase with an explicit state-based monitor.

In the hardware experiments, motion capture provides robot and object poses. A push is declared to deviate — and replanning is triggered from the observed state — when the measured object pose departs from the successor state predicted by EvalSim by more than **0.3 m in position** or  $\pi/4$  **in orientation**, or when the transition/contact checks fail (blocked pre-contact transition, contact loss or stall, unsatisfied geometric task predicate, invalidated WCCG connectivity for the remaining schedule). The pose threshold is deliberately loose: a slightly larger or slightly off-axis push is accepted rather than aborted, because such deviations are routinely recoverable by the subsequent tasks. No onboard force sensing or mass/friction estimation is required.

**Changes in the revised manuscript:** Sec. IV-B1 (Page 7) now defines the pose-based deviation thresholds used by the motion-capture monitor, while Sec. III-E1 lists the additional contact, stall, task-predicate, and connectivity triggers.

**Q:** *Overall, this is a good contribution to the field and a well-written paper. The combination of geometric reasoning, recursive pushing, and physics-informed search is interesting, and the real-world validation strengthens the contribution.*

**A:** We thank the reviewer for the encouraging assessment. The revision leaves the planning framework, the recursive-pushing mechanism and the hardware validation unchanged, and concentrates on the clarifications and the additional evaluations requested in the review. Every number quoted above is reproduced from the per-trial records of the revised runs: the evaluation package accompanying this response contains those records, the resolved configuration files and the complete per-condition tables, and Sec. IV-A1 of the manuscript states the shared protocol (timeout, control-step budget and reachability checks) that applies to PushAround and to both baselines alike. The recursive-pushing mechanism that the reviewer highlights is also the main reason the dense-clutter runs recover once a direct push is blocked; the no-recursion ablation in Sec. IV-A6 quantifies how much of the reported success rate depends on it.

## To Reviewer 4

**Q:** *This paper presents PushAround, a physics-informed hybrid search method for collaborative path clearing. The method uses a W-clearance connectivity graph to identify blocking gaps, ranks these gaps, and generates multi-robot pushing actions that are checked using short-horizon physics simulation. The paper includes simulation comparisons, ablation studies, scalability tests, and hardware experiments.*

*Overall, I think the paper addresses an interesting problem and the proposed framework is practical. The combination of geometric reasoning and physics simulation is useful, and the experimental results are encouraging. I would support acceptance after revision, but I think a few parts of the method need some clarification.*

**A:** We thank the reviewer for the supportive assessment and for pinpointing four places where the presentation was under-specified. Each is addressed below with the corresponding revised text. The supplementary evaluation package has also been updated where relevant.

**Q:** *First, the WCCG connectivity condition is important since it is used throughout the search and also determines termination. The intuition is reasonably clear, but I think the paper would benefit from a little more justification here, or perhaps a comparison with a standard configuration-space connectivity check. This seems especially relevant when the obstacles are non-convex.*

**A:** We agree, and we provide both the requested comparison and an empirical validation. Standard alternatives first inflate the obstacles by  $W/2$  and then test connectivity on an occupancy grid (BFS/A\* or a distance transform) or on a sampling-based roadmap (PRM/RRT-style). These approaches approximate the answer at a chosen grid resolution or sample density and generally return a binary connectivity result; the representation must also be updated as obstacles move. WCCG instead builds an embedded graph directly from the clearance-relevant geometry. Its annotated bridges expose the separating bottleneck and frontier sequence that PIHS uses to choose the next obstacle, in addition to answering the connectivity query. The same-face test is the graph query used to evaluate the terminal condition; its validity follows from the  $W/2$  inflation interpretation under the stated decomposition. Non-convex obstacles are handled by the clearance-relevant convex decomposition described in Sec. III-A2 (Page 3).

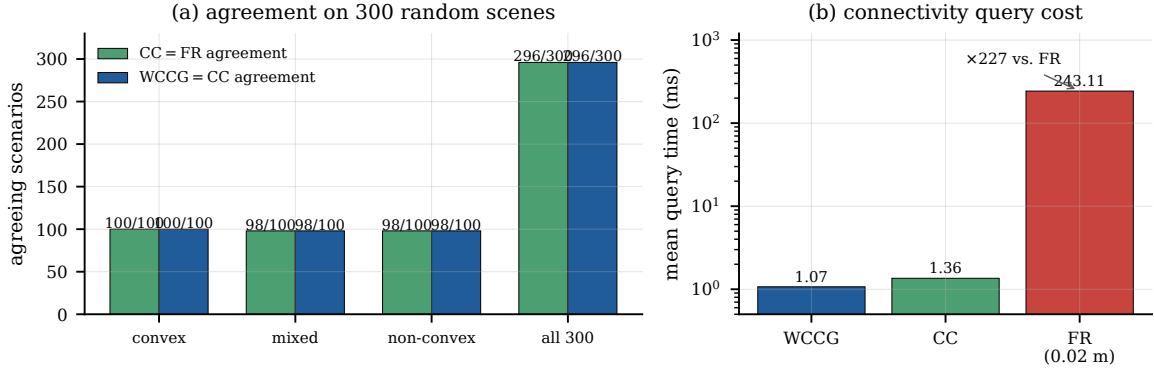
Because geometric degeneracies and decomposition choices still matter in practice, we compare WCCG against two independent references on 300 random scenes stratified into convex, mixed, and non-convex obstacles: a continuous configuration-space (CC) reference and a rasterized reference (FR) at 0.01 m oracle resolution, with a coarser 0.02 m raster used only as a stability check. CC and FR disagree in **4/300** cases, all of which are raster-unstable near narrow or tangential contacts. On all 300 scenes, WCCG agrees with the continuous reference and produces **zero false positives and zero false negatives**; a false positive would be the dangerous case because it could terminate PIHS prematurely. WCCG answers a query in **1.07 ms** on average, compared with 1.36 ms for CC and 243.1 ms for FR. The latter ratio is an implementation-level **227 $\times$**  speed difference, and WCCG additionally supplies the gap graph consumed by PIHS.

**Changes in the revised manuscript.** Sec. III-A and Sec. III-A2 (Page 3) now identify occupancy-grid and sampling-based connectivity checks as the main alternatives and explain why WCCG exposes the bottleneck structure needed by PIHS. Sec. IV-A3 and Table I (Page 6) add the 300-scene comparison and the timing scope:

- The manuscript now derives the criterion from configuration-space inflation, describes the clearance-relevant treatment of non-convex obstacles, and states that consistency with CC and

**Table 7:** WCCG connectivity validation on 300 random scenes (10 movable obstacles,  $8 \times 8$  m, clearance  $W=1$  m). FP/FN are WCCG errors with respect to the continuous reference. Times are means over 100 scenes per category.

| Category     | CC=FR          | WCCG vs. CC    |          |          | Time (ms)                         |                                    |                                   | FR/WCCG    |
|--------------|----------------|----------------|----------|----------|-----------------------------------|------------------------------------|-----------------------------------|------------|
|              |                | agree          | FP       | FN       | CC                                | FR                                 | WCCG                              |            |
| Convex       | 100/100        | 100/100        | 0        | 0        | $1.23 \pm 0.13$                   | $282.9 \pm 52.3$                   | <b><math>0.86 \pm 0.15</math></b> | 330        |
| Mixed        | 98/100         | 100/100        | 0        | 0        | $1.33 \pm 0.13$                   | $242.4 \pm 55.7$                   | <b><math>1.05 \pm 0.17</math></b> | 231        |
| Non-convex   | 98/100         | 100/100        | 0        | 0        | $1.51 \pm 0.18$                   | $204.1 \pm 51.4$                   | <b><math>1.30 \pm 0.19</math></b> | 156        |
| <b>Total</b> | <b>296/300</b> | <b>300/300</b> | <b>0</b> | <b>0</b> | <b><math>1.36 \pm 0.19</math></b> | <b><math>243.1 \pm 62.0</math></b> | <b><math>1.07 \pm 0.25</math></b> | <b>227</b> |



**Figure 6:** WCCG validation. (a) Agreement per category: CC and FR differ on 4 of 300 scenes (all raster-unstable), whereas WCCG matches the continuous reference on all 300 with zero false positives/negatives. (b) Mean connectivity-query cost on a logarithmic scale.

FR is evaluated numerically.

The criterion follows the standard configuration-space inflation interpretation: expanding obstacles by a disk of radius  $W/2$  turns gaps narrower than  $W$  into overlapping barriers. Non-convex obstacles are handled through clearance-relevant decomposition, after which WCCG records the separating bottlenecks.

- Table I reports agreement, false-positive/false-negative counts, and runtime on the same 300 scenes; its note specifies that the timing comparison uses one harness and a compiled WCCG backend.
- The text now says that WCCG avoids introducing a workspace-grid resolution into repeated queries, while still depending on a correct decomposition and bridge construction.

Conventional clearance checks either rasterize the workspace into an inflated occupancy grid or query sampling-based roadmaps, and must therefore fix a resolution and re-query the map whenever an obstacle moves. Built directly from obstacle geometry, the WCCG introduces no grid-resolution parameter in these repeated queries.

**Q:** Second, I was a little confused about the relationship between Eq. (5) and the actual search procedure. Eq. (5) is written as an optimization problem, while the implementation uses sampled pushing directions, limited lookahead, and a heuristic best-first search that terminates once a feasible solution is found. The authors should clarify whether any optimality or completeness claim is intended. If not, it may be better to describe Eq. (5) as the objective used to guide the search.

**A:** We agree with the reviewer’s suggested framing. Eq. (5) is now introduced as the clearance cost  $\mathcal{J}_{\text{clr}}(\pi)$  that ranks candidate schedules. PIHS is a feasibility-oriented heuristic best-first search over a finite sampled set of pushing directions and contact modes; it uses bounded lookahead and stops at the first simulator-validated  $W$ -clear solution. The remaining-cost term is not required to be admissible, so the first feasible schedule is not claimed to minimize  $\mathcal{J}_{\text{clr}}$ .

Completeness is likewise not guaranteed. A feasible scene may require an unsampled direction, contact assignment, or deeper recursion, or may exceed the finite computation budget. The returned schedule is guaranteed only to satisfy the terminal clearance condition and the validated transitions under the evaluation model. The empirical node-expansion counts and ablations describe search efficiency and solution-finding reliability; they do not constitute a formal solution-quality guarantee. These distinctions are now stated in Secs. II-D and III-D (Pages 2 and 5).

**Q:** *Third, it is not completely clear how the optimized contact forces are used on the hardware. The pushing mode includes both contact points and forces, but the execution section mainly describes moving the robots to the contact configuration and then following the desired pushing direction. It would be useful to briefly explain the low-level controller and whether these forces are actually commanded or are mainly used to check feasibility. Related to this, I am also not sure whether coupled motion between multiple objects, such as in the second hardware experiment, is explicitly considered during planning or is mostly handled by the physics simulation.*

**A:** Thank you for raising both points. The optimized forces are not commanded on the hardware; they are used to certify wrench feasibility, while the hardware tracks a short-horizon object-motion reference generated by the simulator.

For a candidate contact tuple, the LP checks whether the robots can generate a wrench compatible with the desired object motion within the modeled force, friction, and unilateral-contact constraints. The optimized force vector is therefore a feasibility and ranking variable, not a hardware setpoint. Over the next control interval, EvalSim predicts a short forward reference (a few centimeters and a few degrees); the velocity-level controller converts this reference and the motion-capture pose error into planar velocity commands for the Mecanum UGVs. Feedback and replanning absorb the difference between the reference and the realized motion.

Coupled motion is handled by the rollout. Although a high-level strategy names one directly targeted obstacle, EvalSim advances all robots and all movable bodies jointly; object–object contacts are resolved by the physics engine and the complete successor scene is returned to the search. The adjacent T-shaped block in hardware Scenario 2 therefore moves as a contact-induced outcome rather than as a separately scripted task.

**Changes in the revised manuscript.** The manuscript now clarifies, in Secs. II-C, III-C3, III-E1, and IV-B1–B2 (Pages 2, 4–7), that the optimized force is a feasibility variable, the controller tracks a short-horizon simulator reference using pose feedback, and coupled object motion is resolved by the joint rollout:

- The force-feasibility, velocity-control, motion-capture, and joint-rollout interfaces are now described explicitly in the corresponding sections.

*The wrench feasibility encoded in  $\xi$  acts as a screening condition rather than a force setpoint: it certifies that the chosen contact configuration can realize the required object motion within the robots’ actuation limits. (Sec. III-C) and a mode is retained only if its object wrench lies within the robots’ actuation limits, and the controller tracks the planned contact poses and pushing velocities with feedback. Planning therefore runs in simulation through EvalSim, and every executed push returns an observed state that re-enters the planner. (Sec. III-E1)*

- The hardware discussion now states that the neighboring-object motion in Scenario 2 is an emergent outcome of EvalSim and is fed back through the updated scene and WCCG.

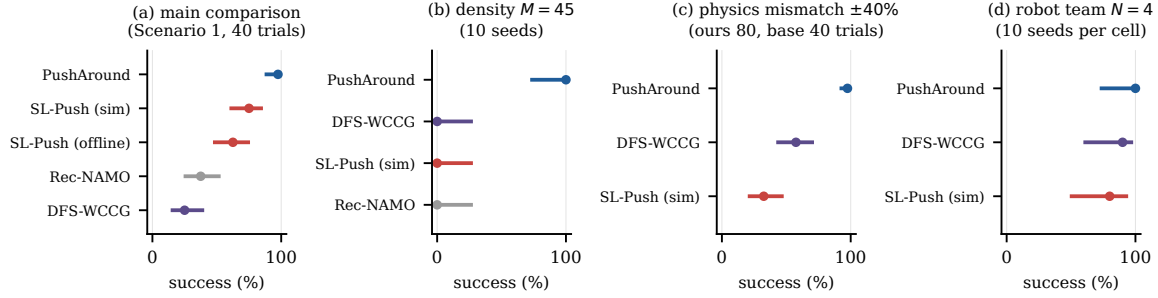
The two Mecanum UGVs are velocity-controlled, and the optimized contact forces are used only for feasibility evaluation rather than direct motor control. A push triggers replanning only when the measured object position error exceeds 0.3 m or its orientation error exceeds  $\pi/4$ . (Sec. IV-B1) and This neighboring-object motion is not a separately encoded action but an emergent outcome of the coupled rigid-body rollout in EvalSim, whose snapshot is used to update the WCCG and continue replanning. (Sec. IV-B2)

**Q:** *The experiments are generally convincing. The paper already mentions that the baselines use the same W-clearance criterion, object distribution, and execution simulator. However, a few more implementation details, such as robot reachability checks, timeout limits, and computation budget, would make the comparison easier to interpret. I would also suggest adding error bars or standard deviations, particularly for the scalability experiment where only five seeds are used for each obstacle density.*

**A:** We have made the protocol explicit and now report dispersion throughout. The full protocol is summarized in Table 8.

- A contact candidate is retained only if the associated pre-contact pose is collision-free and the transition planner can reach it from the current robot configuration; a multi-robot tuple is rejected if any assigned robot fails this check. All compared methods use this same check, together with the same footprints, obstacle realizations, success criterion, and execution simulator.
- All suites use a per-trial timeout of **800 s**, a per-trial cap of **20,000** simulator control steps, and **8** parallel workers; plan failures, timeouts and timeout-pruned trials are assigned the 800 s value before averaging planning times, and the failure composition is reported per condition.
- The density sweep, the robot-team study (all policies) and the mismatch study for PushAround use 10 seeds per cell; only the *mismatch* baseline supplements use 5 seeds and are labeled as such. Fig. 7 reports Wilson 95% intervals for every headline number, and Table III in the manuscript reports  $\text{mean} \pm \text{s.d.}$
- The original manuscript stated a 400 s cap. The stored logs and the revised protocol use 800 s; the earlier statement and the corresponding failure-time imputation have been corrected.

**Changes in the revised manuscript:** Sec. IV-A1 (Page 6) now specifies the common footprints, scene configurations, collision and reachability checks, simulator, 800 s timeout, and 20,000-step simulator cap. It also states how unsuccessful trials enter the capped planning-time statistics.



**Figure 7:** Wilson 95% confidence intervals of the headline success rates. The intervals make sampling uncertainty explicit: the density and mismatch comparisons show clear separation, whereas some robot-team cells overlap and are interpreted descriptively.

**Table 8:** Protocol and budget summary for all reported studies. All suites share the simulator, the success criterion, and the collision/reachability checks.

| Study                        | Workspace      | $M$   | $N$ | Seeds                    | Timeout / cap | Workers |
|------------------------------|----------------|-------|-----|--------------------------|---------------|---------|
| Main comparison (Table II)   | $8 \times 5$ m | 10    | 2   | 40 trials                | 800 s / 20k   | 8       |
| Density (Table III, Fig. 10) | $8 \times 8$ m | 15–45 | 2   | 10 per density           | 800 s / 20k   | 8       |
| Mismatch (Table III)         | $8 \times 8$ m | 30    | 2   | 10 (ours) / 5 (baseline) | 800 s / 20k   | 8       |
| Robot scaling (Table III)    | $8 \times 8$ m | 30    | 2–4 | 10 per cell              | 800 s / 20k   | 8       |
| WCCG validation (Table I)    | $8 \times 8$ m | 10    | —   | 300 scenes               | n/a           | 4 (C++) |

Scene generation in the generalization studies uses movable-object masses sampled from  $[20, 30]$  kg; the Scenario-1 comparison uses  $[5, 15]$  kg. PushAround’s density results are produced with a precomputed mode table (`use_mode_table`), whereas the mismatch and robot-scaling studies use greedy mode optimization; this explains the small planning-time differences between the corresponding nominal rows (e.g. 11.7 s versus 16.8 s at  $M=30$ ,  $N=2$ ) and does not affect success rates.

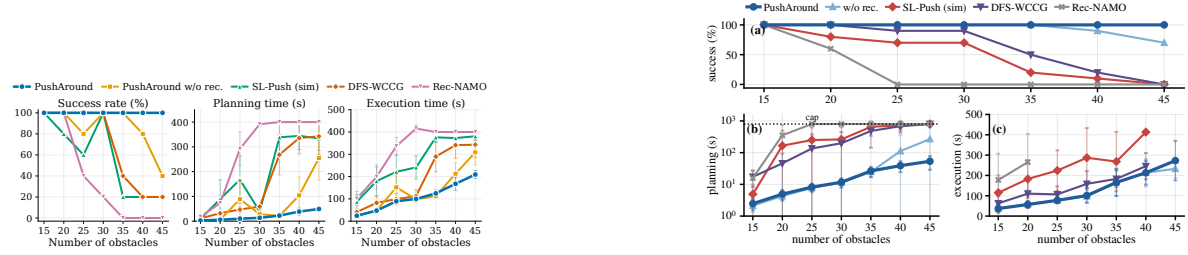
**Q:** Overall, I think the approach is technically reasonable and the experiments support the proposed method. My comments are mainly about clarification rather than major technical concerns, and I would support acceptance after these revisions.

**A:** We thank the reviewer for the supportive assessment. The four clarifications above are addressed without changing the method, and the added protocol details, dispersion statistics and the configuration-space comparison are intended to make the evaluation easier to interpret. The figures that changed in this revision are reproduced side by side with their original versions in the following section, so that all four changes can be checked at a glance, and the complete data tables for each study are placed next to the reply they support rather than in an appendix.

## Updated and added figures

Three figures of the manuscript changed in this revision. In each pair the left image is the original version (first submission) and the right image is the revised version; the page refers to the location in the revised manuscript.

### F1. Obstacle-density scaling: revised Fig. 10 (Page 7)



(a) Original: five seeds per density, mean only, 400 s cap.

(b) Revised: ten seeds per density, mean  $\pm$  s.d., 800 s cap.

**Figure 8:** Obstacle-density scaling, original (left) versus revised (right). The panel structure is unchanged, so the two images are directly comparable: the revision doubles the seeds per density, adds dispersion to the timing panels, and uses the actual 800 s timeout instead of the 400 s cap of the summary.

### F2. Simulation snapshots: revised Fig. 7 (Page 6)

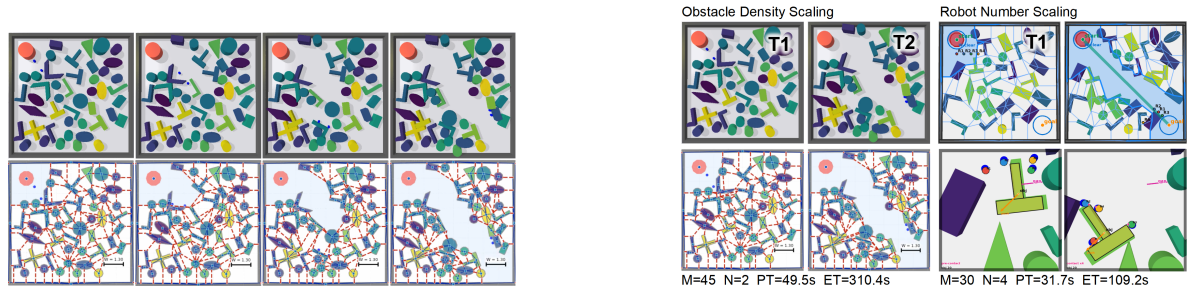


(a) Original: one overlay inset did not match its snapshot.

(b) Revised: regenerated with the correct inset.

**Figure 9:** Execution and planning alignment in the two simulation scenarios, original (left) versus revised (right). One inset of the original composite was incorrect: the fourth sub-image of its bottom row duplicated the sixth, so that WCCG overlay did not match the snapshot above it. The panel has been regenerated; the caption and the message are unchanged.

### F3. Scalability scenes: revised Fig. 9 (Page 7)



(a) Original: obstacle-density run only.

(b) Revised: density run (left half) plus the four-robot study (right half).

**Figure 10:** Simulation scenes of the scalability evaluation, original (left) versus revised (right). The revised figure covers both axes instead of density alone: the left half keeps the representative  $M=45$ , two-robot run (snapshots above, WCCG overlays below), and the right half adds the four-robot study ( $M=30$ ,  $N=4$ ) with initial and final states above and two magnified views of one pushing task below, illustrating the team-size results of Table III (Page 7).

# PushAround: Collaborative Path Clearing via Physics-Informed Hybrid Search

**Abstract**—The passage of large vehicles in cluttered environments is often blocked by movable obstacles, motivating the use of mobile robot teams to proactively clear traversable corridors. Existing approaches to Navigation Among Movable Obstacles (NAMO) mainly plan abstract sequences of obstacle displacements but often neglect physical feasibility, overlooking robot dimensions, obstacle mass, contact geometry, and required forces, which can lead to strategies that fail in execution. This work introduces *PushAround*, a physics-informed hybrid search framework that incrementally constructs executable clearing plans by coupling geometric connectivity reasoning with short-horizon simulation. Repeated W-clearance connectivity graph (WCCG) queries expose frontier gaps where the vehicle corridor must be widened and rank them by estimated downstream costs, while the search generates direct or recursive pushing candidates – choosing which obstacles to displace, in which directions, and with which contact points and forces – admitting only physics-validated candidates. **Additional validation characterizes connectivity-query consistency and sensitivity to physics-model mismatch.** The framework is validated by extensive simulations and hardware experiments.

## I. INTRODUCTION

In many real-world settings, the passage of large vehicles or transport units is obstructed by movable obstacles such as pallets, boxes, or equipment, motivating the deployment of mobile robot teams to actively clear traversable corridors and enable safe passage. This capability is critical in cluttered and unstructured environments such as warehouses, disaster sites, and dense urban spaces. Conventional path planning methods assume static obstacles and thus fail once direct routes are blocked [1]. Navigation Among Movable Obstacles (NAMO) has been studied as a means of creating new routes by pushing, pulling, or rotating obstacles [2], yet most methods abstract away physical feasibility. Factors such as robot dimensions, obstacle size and mass, contact geometry, applied forces and pushing dynamics are typically ignored, producing plans that are *geometrically valid but physically unrealizable in practice*. As shown in Fig. 1, this difficulty is amplified when a large robot cannot reach obstacle boundaries, while a small robot lacks sufficient pushing force. Cooperative pushing by several small robots is a viable alternative, but introduces strong physical coupling since simultaneous pushes can trigger chained motions of multiple obstacles, making prediction and planning substantially more complex.

### A. Related Work

NAMO extends motion planning to cluttered environments by planning obstacle displacements through pushing, pulling, or rotating [2], [3]. Later work considers heuristic search, learning-based local planning, physics-based rollouts, and collaborative pushing [3]–[7]. However, many NAMO formulations simplify robot geometry, actuation limits, object mass, and contact dynamics, which can yield unrealizable plans.



Fig. 1. Collaborative path clearing in both simulation (**Top**) and hardware (**Bottom**). Two robots actively push movable obstacles to create a W-clear path for a large vehicle to traverse from start to goal, via a hybrid search over the sequence of obstacles to push and the associated pushing modes.

Collaborative pushing has been widely studied in multi-robot manipulation: foundational work on the mechanics of pushing and limit surfaces [8], [9], cooperative transport with robot swarms [10], multi-robot box pushing with reinforcement learning [11], collaborative pushing of polytopic objects in cluttered scenes [7], and learning-based locomanipulation [12]. These studies focus on a prescribed object rather than multi-object corridor clearing.

Physics-informed planning incorporates dynamics models, contact reasoning, or simulation feedback into motion generation: learned dynamics for humanoid contact planning [13], multi-contact force control [14], demonstration-guided optimal control over contact modes [15], physics simulation interleaved with multi-agent pathfinding [16], [17], and learned physical priors [18], [19]. **These methods nonetheless assume a prescribed manipulation target or few objects, and physical realizability is usually argued rather than validated before execution.** Corridor clearing differs on both counts: many movable obstacles are rearranged jointly, and the cleared configuration must remain traversable for a large vehicle.

### B. Our Method

As shown in Fig. 2, this work introduces **PushAround**, a physics-informed framework for collaborative multi-robot pushing that actively constructs traversable corridors for large vehicles in cluttered environments. Its central novelty is a hybrid search that jointly determines which obstacles to displace and how to push them, coupling the high-level sequence of obstacle clearing with the low-level pushing modes that specify contact points, directions, and forces. The framework integrates three components: a W-clearance connectivity graph (WCCG) that certifies vehicle passage under width constraints, a gap-ranking strategy that prioritizes frontier gaps by estimated effort, and pushing-mode generation under geometric constraints. These are encapsulated in a search that incre-

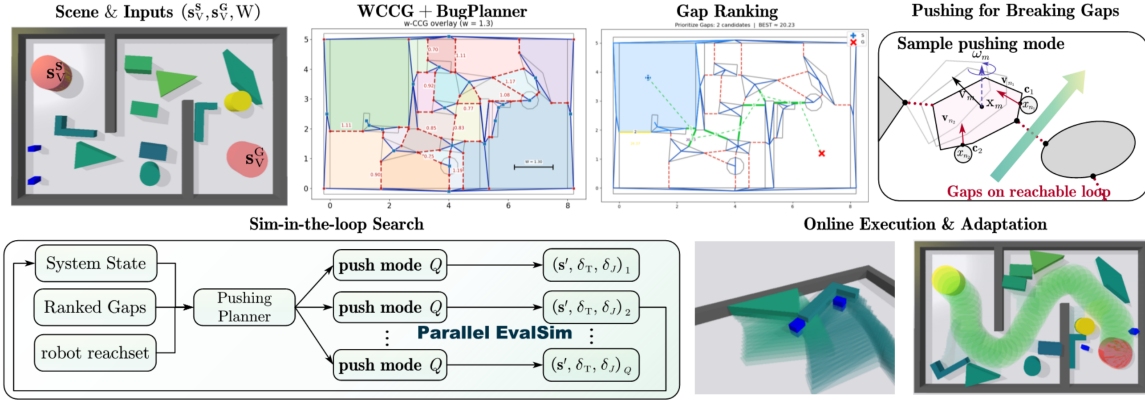


Fig. 2. Overview of the proposed framework. **Top:** Scene inputs, WCCG construction, gap ranking, and sampling of pushing modes for breaking frontier gaps; **Bottom-left:** Sim-in-the-loop hybrid search with parallel validation in simulation; **Bottom-right:** Online execution and adaptation during path clearing.

mentally expands candidate clearing plans while validating their physical realizability, so the resulting plans are not only geometrically valid but also feasible under robot dimensions, object mass, contact geometry, and coupled pushing dynamics. *Since physics validation depends on model fidelity, robustness under execution-model mismatch is also evaluated.*

The contributions are twofold: (I) it introduces a unified framework that couples collaborative pushing with physics-informed verification, producing executable plans for path clearing in cluttered environments; and (II) it shows significant improvements in feasibility, efficiency and scalability over existing NAMO and collaborative clearing methods.

## II. PROBLEM DESCRIPTION

### A. Workspace and Robots

The workspace is a bounded planar region  $\mathcal{W} \subset \mathbb{R}^2$  containing immovable structures  $\mathcal{O} \subset \mathcal{W}$  and  $M \geq 0$  movable rigid polygons  $\Omega \triangleq \{\Omega_1, \dots, \Omega_M\} \subset \mathcal{W}$ . Each movable obstacle  $\Omega_m$  has mass  $M_m$ , inertia  $I_m$ , frictional parameters and state  $\mathbf{s}_m(t) \triangleq (\mathbf{x}_m(t), \psi_m(t))$ , with  $\mathbf{x}_m(t) \in \mathbb{R}^2$  the centroid position and  $\psi_m(t) \in \mathbb{R}$  the orientation;  $\Omega_m(t)$  denotes the region it occupies at time  $t$ . A team of  $N$  robots,  $\mathcal{R} \triangleq \{R_1, \dots, R_N\}$ , operates as a cooperative unit. Each robot  $R_i$  is a rigid body with state  $\mathbf{s}_{R_i}(t) \triangleq (\mathbf{x}_{R_i}(t), \psi_{R_i}(t))$  and footprint  $R_i(t)$ . The instantaneous free space is given by:

$$\widehat{\mathcal{W}}(t) \triangleq \mathcal{W} \setminus (\mathcal{O} \cup \{\Omega_m(t)\}_{m=1}^M \cup \{R_i(t)\}_{i=1}^N), \quad (1)$$

where  $\widehat{\mathcal{W}}(t)$  excludes regions occupied by immovable obstacles, movable obstacles, and robots.

### B. External Vehicle and Clearance Goal

An external vehicle  $\mathcal{V}$  whose footprint is enclosed by a disk of radius  $W/2 > 0$  must navigate within the workspace from a start position  $\mathbf{x}_V^s \in \mathcal{W}$  to a goal position  $\mathbf{x}_V^g \in \mathcal{W}$ . Since movable obstacles may obstruct the way, the clearance goal is to rearrange them so that the workspace contains a passage from  $\mathbf{x}_V^s$  to  $\mathbf{x}_V^g$  that can accommodate this enclosing disk, i.e., a passage of width at least  $W$ . At time  $t$ , this is expressed by the admissible reference-center space  $\mathcal{F}_W(t) \triangleq \{p \in \widehat{\mathcal{W}}(t) \mid \mathbb{B}_{W/2}(p) \subset \widehat{\mathcal{W}}(t)\}$ , where  $\mathbb{B}_{W/2}(p)$  is the disk of radius  $W/2$  centered at  $p$ . The  $W$ -clearance condition is then given by:

$$\exists \mathcal{P}_V^W \subset \mathcal{F}_W(t) : \mathbf{x}_V^s \rightsquigarrow \mathbf{x}_V^g, \quad (2)$$

which certifies a collision-free centerline path for any vehicle footprint contained in the enclosing disk.

### C. Collaborative Pushing Modes

To enlarge blocked gaps, the robots may actively reconfigure  $\Omega$  by pushing movable obstacles. For obstacle  $\Omega_m \in \Omega$ , a pushing mode is denoted by  $\xi_m \triangleq (\mathbf{c}_m, \mathbf{u}_m)$ , where  $\mathbf{c}_m = (c_1, \dots, c_N)$  is an ordered tuple of contact points on  $\partial\Omega_m$ , and the ordering specifies the robot-to-contact assignment. The vector  $\mathbf{u}_m \in \mathbb{R}^{2N}$  stacks the corresponding planar contact forces, expressed in the body frame of  $\Omega_m$ . For a contact tuple  $\mathbf{c}_m$ , the set of admissible forces  $U_m(\mathbf{c}_m) \subset \mathbb{R}^{2N}$  collects contact forces satisfying the robot pushing limits, robot-object friction, and unilateral contact constraints. The body wrench induced by a mode is defined as:

$$\mathbf{q}_m(\xi_m) \triangleq G_m(\mathbf{c}_m)\mathbf{u}_m, \quad (3)$$

where  $G_m(\mathbf{c}_m) \in \mathbb{R}^{3 \times 2N}$  is the contact-to-wrench mapping; and  $\mathbf{q}_m(\xi_m) \in \mathbb{R}^3$ . The admissible mode set is state dependent as denoted by  $\Xi_m(\mathbf{s})$ . It contains modes whose contacts are reachable from the current robot configurations, pre-contact configurations are feasible in the current clutter, and forces satisfy  $\mathbf{u}_m \in U_m(\mathbf{c}_m)$ . Hence, applying a feasible mode  $\xi_m \in \Xi_m(\mathbf{s}(t))$  yields the physical transition:

$$\mathbf{s}(t^+) \triangleq \Phi(\mathbf{s}(t), m, \xi_m), \quad (4)$$

where  $\mathbf{s}(t)$  stacks all robot and obstacle states. *The force vector  $\mathbf{u}_m$  serves as a feasibility variable to certify wrench generation under the modeled contact and actuation constraints rather than a directly commanded control input.*

### D. Problem Statement

The goal is to compute a *hybrid schedule* that reconfigures movable obstacles to admit a  $W$ -feasible path for the external vehicle. The schedule is defined as:

$$\pi \triangleq \{(\Omega_{m_k}, \xi_k, \Delta t_k)\}_{k=1}^K, \quad (5)$$

where  $\Omega_{m_k} \in \Omega$  is the manipulated obstacle,  $\xi_k \in \Xi_{m_k}$  is its pushing mode, and  $\Delta t_k > 0$  is the execution duration. The search is guided by

$$\mathcal{J}_{\text{clr}}(\pi) \triangleq T + \alpha \sum_{k=1}^K J(m_k, \xi_k, \mathbf{s}(t_k)), \quad (6)$$

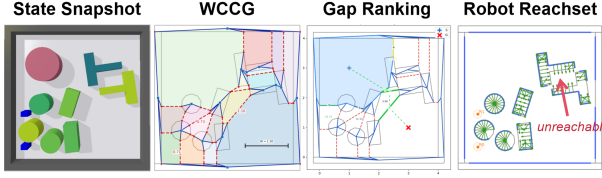


Fig. 3. Core components of the physics-informed hybrid search: WCCG construction, gap ranking, mode generation, and physics validation.

where  $T \triangleq \sum_{k=1}^K \Delta t_k$  denotes the robotic clearance duration excluding subsequent vehicle traversal, and  $J(\cdot)$  is the simulation-estimated manipulation effort. Vehicle traversal is considered only through the terminal  $W$ -clearance condition (2). The constraints (1)–(4) enforce valid pushing transitions. Equation (6) ranks feasible schedules; PIHS uses finite sampling and does not guarantee global optimality of the continuous hybrid objective.

**Remark 1.** The objective couples obstacle selection, pushing-mode generation and switching into one hybrid planning problem, yielding a combinatorial-continuous search space richer than classical NAMO formulations with hand-crafted contact modes [8], [10], [11]. ■

### III. PROPOSED SOLUTION

As shown in Fig. 2, PushAround first uses the  $W$ -Clearance Connectivity Graph in Sec. III-A and gap ranking in Sec. III-B to identify and prioritize blocking gaps. It then generates and evaluates physically feasible collaborative pushing modes in Sec. III-C. These components are integrated into the simulation-in-the-loop hybrid search in Sec. III-D, followed by the online execution procedure summarized in Sec. III-E.

#### A. $W$ -Clearance Connectivity Graph

The feasibility of routing the vehicle reduces to whether a  $W$ -width corridor connects  $\mathbf{x}_V^S$  and  $\mathbf{x}_V^G$ . A  $W$ -clearance connectivity graph (WCCG) is introduced here to encode obstacle adjacencies under clearance  $W$  and enable fast connectivity queries. Conventional clearance checks either rasterize the workspace into an inflated occupancy grid or query sampling-based roadmaps, and must therefore fix a resolution and re-query the map whenever an obstacle moves [20]. Built directly from obstacle geometry, the WCCG introduces no grid-resolution parameter in the repeated connectivity queries used during planning.

1) *Graph Construction:* The WCCG is built by decomposing all clearance-relevant obstacle and workspace boundaries into convex components or segments. From each component  $C$ , a centroid node  $v_c$  is created. When two components  $C_u$  and  $C_v$  have closest points  $p_u \in C_u$  and  $p_v \in C_v$  with distance less than  $W$ , bridge nodes are added at  $p_u$  and  $p_v$ . These bridge nodes are connected by a bridge-bridge edge, annotated with the gap width  $w_{uv} \triangleq \|p_u - p_v\|$ , and further linked back to their centroids with centroid-bridge edges. Narrow passages with  $w_{uv} < W$  are marked as potential bottlenecks. The resulting graph is given by:

$$\mathcal{G}_W \triangleq (\mathcal{V}, \mathcal{E}_c \cup \mathcal{E}_b), \quad (7)$$

where  $\mathcal{V}$  are centroid and bridge nodes;  $\mathcal{E}_c$  contains centroid-bridge edges; and  $\mathcal{E}_b$  for bridge-bridge edges with widths  $w_{uv}$ .

2) *Connectivity Criterion:* Once  $\mathcal{G}_W$  has been constructed, connectivity queries can be performed without explicitly computing a geometric path. Let  $\text{Face}(\mathcal{G}_W)$  denote the set of connected planar faces induced by the embedded WCCG, i.e., the maximal regions separated by obstacle components and WCCG bottleneck edges. The criterion for (2) is given by:

$$(\mathbf{x}_V^S, \mathbf{x}_V^G \in \mathcal{F}_W(t)) \wedge (\exists F \in \text{Face}(\mathcal{G}_W) : \mathbf{x}_V^S, \mathbf{x}_V^G \in F) \quad (8)$$

The criterion follows the standard configuration-space inflation interpretation: expanding obstacles by a disk of radius  $W/2$  turns gaps narrower than  $W$  into overlapping barriers. Non-convex obstacles are handled through clearance-relevant decomposition, after which WCCG records the separating bottlenecks. The consistency between WCCG connectivity and continuous and rasterized configuration-space references is further evaluated numerically in Sec. IV-A3. A frontier-tracing procedure, similar to the BugPlanner [21], is then used to extract a skeleton  $\Sigma$  as an ordered sequence of centroid and bridge nodes. Starting from  $\mathbf{x}_V^S$ , it casts a ray toward  $\mathbf{x}_V^G$  and follows the encountered frontier loop until either a valid exit is found or a blocking cycle is returned.

#### B. Ranking of Frontier Gaps

When the condition in (8) fails, the vehicle cannot reach  $\mathbf{x}_V^G$  from  $\mathbf{x}_V^S$  through a  $W$ -clear path  $\mathcal{P}_V^W$ . In this case, the planner must prioritize *blocking gaps* on the reachable frontier of  $\mathcal{G}_W$  according to their predicted downstream cost, and feed the ranked list to the hybrid search as local clearance targets.

1) *Frontier Extraction and Candidate Gaps:* A BugPlanner query on  $\mathcal{G}_W$  above either confirms connectivity or returns a counter-clockwise frontier loop  $\mathcal{L}$  that separates the two endpoints. The bridge-bridge edges on  $\mathcal{L}$  define the first-hop candidate set  $\Gamma_{\mathcal{L}} \triangleq \{g_1, \dots, g_K\}$ , where each gap  $g_k$  is represented by its two closest boundary points, its width  $w_{g_k} < W$ , and the adjacent obstacles that can potentially be moved. These candidates are then ranked by predicted downstream cost and passed to the hybrid search as local clearance targets.

2) *Evaluation for Immediate Cost:* Each candidate  $g \in \Gamma_{\mathcal{L}}$  is evaluated by combining the cost for the robots to reach the gap and the effort required to widen it. Let  $\mathbf{s}_{\mathcal{R}}$  be the current robot positions, and  $\mathbf{o}_g$  as the outside insertion point of gap  $g$ . The resulting one-hop cost is given by:

$$\mathcal{C}(g \mid \mathcal{L}, \mathbf{s}_{\mathcal{R}}) = \lambda_t \mathcal{C}_t(\mathbf{s}_{\mathcal{R}}, \mathbf{o}_g) + \lambda_p (\mathcal{C}_{dp}(g) + \mathcal{C}_{rp}(g)) \quad (9)$$

where  $\lambda_t, \lambda_p > 0$  weight transition and pushing costs, respectively;  $\mathcal{C}_t$ ,  $\mathcal{C}_{dp}$ , and  $\mathcal{C}_{rp}$  denote transition, direct-pushing, and recursive-pushing costs.

3) *Long-term Cost w.r.t. Goal:* Furthermore, to prioritize gaps closer to the goal, a heuristic is added to the evaluation, i.e.,  $h(g) = \eta \|\mathbf{o}(g) - \mathbf{x}_V^G\|$ , with the scaling constant  $\eta > 0$ . Lastly, a short A\*-style presearch is adopted to cross each candidate gap, recompute the next frontier, and continue for a limited width and depth. The *cost-to-connect* is predicted by:

$$\widehat{\text{Cost}}(g) \triangleq \mathcal{C}(g \mid \mathcal{L}, \mathbf{s}_{\mathcal{R}}) + \sum_{g' \in \Pi^*(g)} \{\mathcal{C}(g' \mid \cdot) + h(g')\}, \quad (10)$$

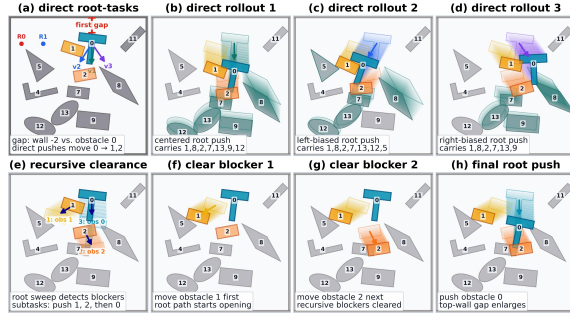


Fig. 4. **Direct pushing** moves a gap-side blocker along a gap-opening direction, whereas **recursive pushing** first clears movable blockers in the swept region and then returns to the root gap-opening task.

where  $\Pi^*(g)$  is the sequence of subsequent gaps discovered after virtually crossing  $g$ . Consequently, the final output of this module is the ranked list of candidate gaps, i.e.,  $\text{Rank}(\Gamma_{\mathcal{L}}) \triangleq \text{argsort}_{g \in \Gamma_{\mathcal{L}}} \widehat{\text{Cost}}(g)$ , in ascending predicted cost. This ordered set is passed to the hybrid search module, such that only the promising gaps are expanded first.

### C. Pushing Mode Generation and Evaluation

For a selected frontier gap  $g$ , this subsection constructs a set of candidate pushing strategies  $\Xi_g \triangleq \{(\Omega_m, \mathbf{v}, \xi)\}$ , where  $\mathbf{v} \triangleq (v_x, v_y, \omega) \in \mathbb{R}^3$  is a short-horizon pushing direction for the movable obstacle  $\Omega_m$  induced by  $g$ , and  $\xi \triangleq (\mathbf{c}_m, \mathbf{u})$  is the associated pushing mode in (4). The wrench feasibility encoded in  $\xi$  acts as a screening condition rather than a force setpoint: it certifies that the chosen contact configuration can realize the required object motion within the robots' actuation limits, after which the candidate is passed to the physics rollout for validation.

1) *Direct and Recursive Pushing Tasks:* For *direct push*, let  $\Omega_m$  denote one of the two movable obstacles adjacent to  $g$ . For each such obstacle, a small set of candidate body velocities  $\{\mathbf{v}_m^k\}_{k=1}^{K_m}$  is sampled with a bias toward enlarging the gap  $g$ , while a few angular components are included to account for local geometry. If the swept region of a direct push intersects movable blockers, *recursive push* tasks are required instead: for each out-layer blocker  $\Omega_m$ , the nominal direction moves it away from both the swept region and the reference gap sequence  $\Pi^*(g)$ . Recursion therefore creates a bounded dependency chain which prioritizes the final blocker  $\Omega_m$ , as shown in Fig. 4.

2) *Pushing Mode Generation:* At a search state  $\mathbf{s}$ , a pushing mode is generated for each candidate direction  $\mathbf{v}_m^k$  and obstacle  $\Omega_m$ . In this subsection, contact tuples, object velocities, forces, and wrenches are expressed in the body frame of  $\Omega_m$ . For a contact tuple  $\mathbf{c}$ , its multi-directional force feasibility is:

$$J_{\text{MF}}^m(\mathbf{c}, \mathbf{v}_m^k) \triangleq \sum_{\mathbf{v}_d \in \mathcal{D}_m^k} w_d \min_{\mathbf{u} \in U_m(\mathbf{c})} \|\mathbf{q}_m(\mathbf{c}, \mathbf{u}) + \mathbf{q}_m^\mu(\mathbf{v}_d)\|, \quad (11)$$

where  $\|\cdot\|$  is a weighted  $\ell_1$  wrench norm, the finite set  $\mathcal{D}_m^k$  perturbs  $\mathbf{v}_m^k$ , and  $\mathbf{q}_m^\mu(\mathbf{v}_d)$  is the quasi-static friction wrench; the inner minimization is a small LP over the admissible forces  $U_m(\mathbf{c})$ . A state- and direction-dependent contact set  $\mathcal{A}_m^k(\mathbf{s}) \subset \partial\Omega_m$  is drawn from the robot-reachable boundary

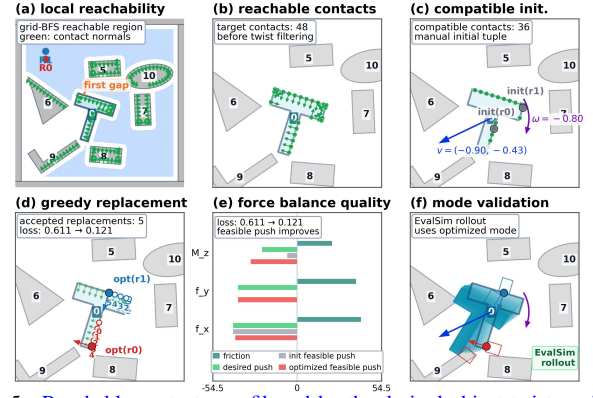


Fig. 5. Reachable contacts are filtered by the desired object twist, assigned to robots, refined by local replacement, and finally validated by simulation.

of  $\Omega_m$  by velocity–force compatibility filtering (Fig. 5), keeping a contact only if its robots reach collision-free pre-contact poses. Starting from cached layouts or random initializations, the tuple is refined by sequential greedy replacement: at the  $\ell$ -th step,  $\mathbf{c}_{r,c}^\ell$  replaces the  $r$ -th contact with  $c$ , and the best replacement is:

$$\mathbf{c}^* = \underset{\mathbf{c} \in \mathcal{A}_m^k(\mathbf{s})}{\text{argmin}} J_{\text{MF}}^m(\mathbf{c}_{r,c}^\ell, \mathbf{v}_m^k), \quad (12)$$

swept over contact slots until  $J_{\text{MF}}^m$  no longer decreases. A bounded backtracking variant removes the robot-order sensitivity of this replacement. If no full-team assignment exists for a task, e.g., on small faces or under the pairwise pusher-clearance constraint, one robot is released for that single push (charged a penalty when ranking modes, so full-team modes are preferred) and the full team is available again afterwards. The best tuple over initializations gives  $\mathbf{c}_m^{k,*}$ , from which  $\mathbf{u}_m^k$  follows from the inner minimization in (11), yielding  $\xi_m^k = (\mathbf{c}_m^{k,*}, \mathbf{u}_m^k)$ ; each mode is cached as a warm start. The best few admissible pairs are kept for gap  $g$  and passed to simulation;  $\mathbf{u}_m^k$  certifies wrench feasibility while hardware tracks the contact geometry from pose/velocity commands.

3) *Pushing Strategy Evaluation:* For a current state  $\mathbf{s}$  and a candidate strategy  $(\Omega_m, \mathbf{v}, \xi) \in \Xi_g$ , a short-horizon rollout evaluates the transition in (4) under the selected mode. As shown in Fig. 5, a strategy is accepted only if the robots reach the pre-contact poses, maintain stable contact, and induce obstacle motion above threshold. The simulation returns:

$$(\mathbf{s}', \delta_T, \delta_J) \triangleq \text{EvalSim}(\mathbf{s}, (\Omega_m, \mathbf{v}, \xi)), \quad (13)$$

where  $\mathbf{s}'$  is the successor state capturing the resulting robot and object configurations;  $\delta_T > 0$  is the time cost; and  $\delta_J$  is the control effort. The rollout jointly simulates all robots and movable bodies, thereby capturing contact-induced object interactions. For efficiency, the pushing rollout may be bypassed when the mode has been validated before and the swept region of  $\Omega_m$  is collision-free.

### D. Physics-Informed Hybrid Search

The proposed physics-informed hybrid search (PIHS) selects blocking gaps and validates corresponding multi-robot pushing strategies through short-horizon simulation, as shown in Fig. 6 and summarized in Alg. 1. Unlike purely geometric

Evaluation of Pushing Modes via Parallel Simulation:

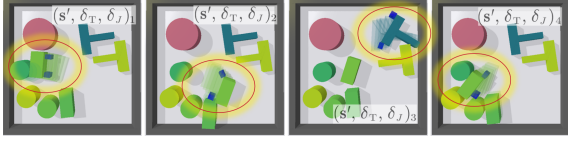


Fig. 6. Sampled push tasks are evaluated via parallel simulation, producing updated states for search expansion.

NAMO planners, PIHS expands only physics-validated strategies through parallel candidate evaluation. It is a feasibility-oriented heuristic search, where Eq. (6) defines the ranking objective for candidate schedules. The search operates on a finite sampled candidate set and therefore does not guarantee global optimality or completeness of the underlying continuous hybrid planning problem.

1) *Tree and Initialization*: The search tree  $\mathcal{T}$  holds nodes  $\nu \triangleq (s, \pi)$ , where  $s$  is the system state of all robots and movables in (4) and  $\pi$  the partial pushing strategy in (6). Two global functions are maintained:  $\text{Rank}(\nu)$  stores the ranked candidate gaps  $\Gamma_{\mathcal{L}_\nu}$  of the frontier loop  $\mathcal{L}_\nu$  together with their exploration costs by (10), and  $\chi(\nu)$  returns the cumulative execution cost from the root to  $\nu$ . The root is  $\nu_0 \triangleq (s_0, \emptyset)$  with initial state  $s_0$ .

2) *Node Selection*: At each iteration, the node with minimum best-first priority is selected from the queue. Priority balances realized cost and estimated remaining effort:

$$f(\nu) \triangleq \chi(\nu) + \min_{g \in \text{Rank}(\nu)} \widehat{\text{Cost}}(g), \quad (14)$$

where  $\chi(\nu)$  is the cumulative execution cost so far, and the second term is the minimum predicted remaining cost among the unexplored gaps of  $\nu$ . The remaining-cost term is a heuristic and is not required to be admissible. The score prioritizes nodes by realized cost and promising future actions, but it does not provide an A\* optimality guarantee.

3) *Node Expansion with Parallel Simulation*: Let the selected node be  $\nu \triangleq (s, \pi)$ . For each ranked gap  $g \in \text{Rank}(\nu)$ , its predicted gap sequence  $\Pi^*(g)$  in (10) is first derived. Then, the mode optimization by (11) provides a candidate strategy set  $\Xi_g \triangleq \{(\Omega_m, \mathbf{v}, \xi)\}$ , including both the direct push of the adjacent movable obstacles, and recursive push of out-layer obstacles. Thus, expansion is prioritized first by the gap ordering in  $\text{Rank}(\nu)$ , and then by the candidate ordering within  $\Xi_g$ . Each candidate strategy  $(\Omega_m, \mathbf{v}, \xi) \in \Xi_g$  is first checked by a geometric quick-pass and, if it passes, evaluated by the short-horizon rollout in (13). A successful evaluation yields  $(s', \delta_T, \delta_J)$ , from which a child node  $\nu' \triangleq (s', \pi')$  is created by appending  $\tau \triangleq (\Omega_m, \mathbf{v}, \xi)$  to the current partial schedule  $\pi$ . The cumulative cost is updated by:  $\chi(\nu') \triangleq \chi(\nu) + \widehat{C}(\nu, \tau)$ , where  $\widehat{C}(\nu, \tau) \triangleq \delta_T + \alpha \delta_J$  is the realized incremental cost. Note that all candidate strategies in the current batch are evaluated *concurrently in parallel*, and a single expansion can produce multiple validated successors.

4) *Termination*: Termination occurs when a node  $\nu$  reaches a state  $s$  for which a  $W$ -clear path  $\mathcal{P}_V^W$  exists from  $\mathbf{x}_V^S$  to  $\mathbf{x}_V^G$  by (8). Thus, the schedule  $\pi$  stored in  $\nu$  is a feasible solution satisfying the terminal clearance condition and the validated pushing transitions. It encodes both the sequence of gaps to

### Algorithm 1: Physics-Informed Hybrid Search

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**Input:**  $s_0, \alpha, \text{EvalSim}(\cdot), W$ -criterion by (8)  
**Output:**  $\pi^*$

```

1  $\nu_0 \leftarrow (s_0, \emptyset), \chi(\nu_0) = 0;$ 
2 Init  $\mathcal{Q}$  by (14);
3 while not terminated do
4    $\nu \leftarrow \mathcal{Q}.\text{pop\_min}()$  by (14);
5   foreach  $g \in \text{Rank}(\nu)$  do
6     Compute  $\Pi^*(g)$ , generate  $\Xi_g$ ;
7     foreach  $(\Omega_m, \mathbf{v}, \xi) \in \Xi_g$  do
8        $\tau = (\Omega_m, \mathbf{v}, \xi)$  // Direct or recur.
9        $(s', \delta_T, \delta_J) \leftarrow \text{EvalSim}(\nu, \tau)$  by (13);
10      if success then
11         $\nu' \leftarrow (s', \pi \cup \{\tau\});$ 
12         $\chi(\nu') \leftarrow \chi(\nu) + \delta_T + \alpha \delta_J;$ 
13        Add  $\nu'$  to  $\mathcal{Q}$ ;
14      if  $s'$  admits  $\mathcal{P}_V^W$  by (8) then
15         $\pi^* \leftarrow \pi',$  break;
16 return  $\pi^*;$ 

```

---

clear and the pushing strategies that realize them, without implying that  $\mathcal{J}_{\text{clr}}(\pi)$  is globally minimal. The objective covers the clearing phase only, since the external vehicle traverses the corridor after clearing completes; traversal time is therefore not part of  $T$ .

**Remark 2.** Efficiency comes from three lightweight heuristics: gap-lookahead reduces unproductive branches, MODE-CACHE warm-starts mode optimization, and the quick-pass rollout bypass skips simple motions whose mode was previously validated, avoiding redundant simulations. ■

### E. Overall Analyses

1) *Execution and Online Adaptation*: The hybrid search returns a pushing schedule  $\pi = \{\tau_1, \dots, \tau_K\}$ , with  $\tau_k \triangleq (\Omega_k, \mathbf{v}_k, \xi_k)$ . The schedule is executed in a receding-horizon manner: for each  $\tau_k$  the robots reach the mode  $\xi_k$  and follow the pushing direction  $\mathbf{v}_k$  from the current state, as in (13). On the hardware the optimized contact forces certify feasibility rather than being commanded: a mode is retained only if its object wrench lies within the robots' actuation limits, and over each control interval  $\text{EvalSim}$  predicts a short forward object-motion reference (the expected displacement and rotation of the object), which the velocity-level controller tracks with motion-capture feedback, and each executed push returns an observed state that re-enters the planner. During transitions, robots are reassigned by preserving the circular order of contact points along the obstacle boundary [7], reducing path crossings under sufficient clearance. After each push the state is updated from observation, the WCCG is rebuilt from the current configuration, and execution continues until a  $W$ -clear path  $\mathcal{P}_V^W$  connects  $\mathbf{x}_V^S$  and  $\mathbf{x}_V^G$ . Replanning is triggered by observed execution discrepancies – failed contact establishment, stalled pushing, or invalidated WCCG connectivity – in which case Alg. 1 is called from the current state to derive the updated schedule.

2) *Computational Complexity*: Given  $M$  movable obstacles, the computational cost of one node expansion in Alg. 1 is dominated by candidate generation and physics validation. The WCCG update and gap ranking incur  $\mathcal{O}(M \log M + |\mathcal{V}| + |\mathcal{E}_c| +$

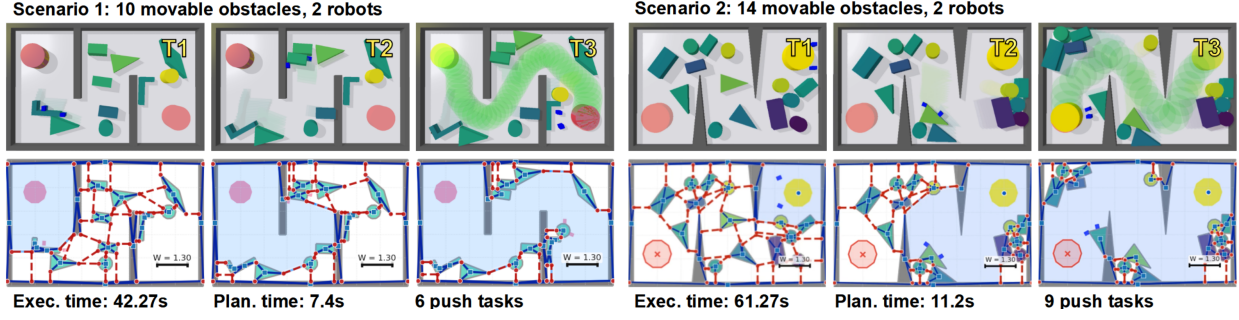


Fig. 7. Execution and planning alignment in Scenario 1 (Left) and Scenario 2 (Right). **Top:** simulated snapshots during path clearing. **Bottom:** corresponding WCCG overlays with the start face in blue. The executed pushes progressively expand the reachable  $W$ -clearance region until the goal is connected.

TABLE I  
WCCG CONNECTIVITY VALIDATION AND MEAN RUNTIME.

| Type       | CC=FR   | FP/FN <sup>†</sup> | Time (ms) |       |             |
|------------|---------|--------------------|-----------|-------|-------------|
|            |         |                    | CC        | FR    | WCCG        |
| Convex     | 100/100 | 0/0                | 1.23      | 282.9 | <b>0.86</b> |
| Non-convex | 98/100  | 0/0                | 1.51      | 204.1 | <b>1.30</b> |
| Mixed      | 98/100  | 0/0                | 1.33      | 242.4 | <b>1.05</b> |
| Total      | 296/300 | <b>0/0</b>         | 1.36      | 243.1 | <b>1.07</b> |

<sup>†</sup>WCCG errors with respect to the continuous configuration-space reference (CC). All runtimes are measured in one harness over the same 300 scenes, with WCCG served by its compiled backend;

$|\mathcal{E}_b|$ ) and  $\mathcal{O}(|\Gamma_{\mathcal{L}_\nu}| \log |\Gamma_{\mathcal{L}_\nu}|)$  costs, respectively. The dominant simulation cost  $\mathcal{O}(\sum_{g \in \text{Rank}(\nu)} |\Xi_g| (C_{\text{LP}} + \rho_\nu T_{\text{sim}}/n_w))$  grows with the number of validated candidates and is reduced by the  $n_w$  parallel workers, so scalability is set by validated candidates rather than by the enumeration of contact assignments. Robot-team size mainly affects mode generation: the greedy stage evaluates at most  $N|A|$  single-contact substitutions instead of the exhaustive  $|A|^N$  assignment, and when it fails, a bounded-backtracking search over robots and candidates expands at most  $B=2000$  partial assignments, a fixed budget independent of the team size; the subset fallback repeats this search for at most one excluded robot. The number of LP feasibility checks per task is therefore at most  $(N|A|+2B)$ , each over the  $2N$  admissible contact forces, rather than growing as  $|A|^N$ .

#### IV. EXPERIMENTS

Numerical experiments are conducted to validate the proposed PIHS algorithm. The algorithms are implemented in Python3 and PyBullet [22] on a computer with an Intel Core i9-14900KF and an NVIDIA RTX 4080. Videos are provided as multimedia attachments.

##### A. Numerical Simulations

1) *Setup and Configuration:* The simulation runs at a step size of  $\Delta t = 1/240$  s with a control period of  $1/40$  s; robots are disk or box pushers consistent with the  $W$ -clearance definition, and movable obstacles have masses sampled from  $[5, 15]$  kg. A trial succeeds when the external vehicle reaches its goal through a  $W$ -clear path. The nominal scenario is an  $8 \times 5$  m workspace with bottleneck-forming bar obstacles, where two robots clear 10–15 movable obstacles under minimum-separation constraints. Unless stated otherwise, comparisons and ablations use Scenario 1 (10 movable obstacles, 40 trials), and Scenario 2 (14 obstacles) provides an additional cluttered

TABLE II  
PERFORMANCE COMPARISON AND ABLATIONS ON SCENARIO 1 OVER 40 RANDOMIZED TRIALS.

| Method / Variant         | Succ.(%)    | PT (s)      | ET (s)      | #Sims        | #Pushes    |
|--------------------------|-------------|-------------|-------------|--------------|------------|
| DFS-WCCG                 | 25.0        | >100.0      | 51.2        | >500         | 8.0        |
| SL-Push (offline)        | 62.5        | 0.06        | 44.1        | 0.0          | 7.3        |
| SL-Push (sim)            | 75.0        | 18.2        | 64.6        | 20.0         | 10.2       |
| Rec-NAMO                 | 37.5        | 13.3        | 42.4        | 0.0          | 7.8        |
| <b>PushAround (ours)</b> | <b>97.5</b> | <b>10.3</b> | <b>28.6</b> | <b>121.5</b> | <b>6.0</b> |
| w/o gap-lookahead        | 82.5        | 16.9        | 35.2        | 178.3        | 7.2        |
| w/o mode-cache           | 92.5        | 17.1        | 33.4        | 164.6        | 6.5        |
| w/o quick-pass           | 90.0        | 23.6        | 41.9        | 208.2        | 6.1        |
| w/o reinsertion          | 85.0        | 12.8        | 31.1        | 151.7        | 7.8        |

case. All methods share the robot footprints, obstacle configurations, collision and reachability checks, and execution simulator, with the same per-trial timeout of 800 s and simulator cap of 20,000 control steps; unsuccessful trials are assigned the 800 s cap in the planning-time statistics.

2) *Results:* Two representative simulation runs are shown in Fig. 7. In Scenario 1 the start and goal lie in different  $W$ -clearance faces, so the planner clears the blocked passage with 6 pushes in 7.4 s and execution finishes in 42.3 s. Scenario 2 requires a 9-push plan in 11.2 s and 61.3 s of execution. In both cases the executed pushes progressively enlarge the reachable  $W$ -clearance region until a complete  $W$ -clear path is found, as shown by the expanding blue start face in the WCCG overlays.

3) *WCCG Connectivity Validation:* WCCG is compared against two references on 300 random scenes covering convex, non-convex and mixed obstacles (Table I). The continuous reference (CC) inflates obstacle and workspace boundaries by  $W/2$  and queries connectivity with a fine continuous collision check at 0.01 m oracle resolution instead of a fixed grid [23]; the rasterized reference (FR) discretizes the same workspace at 0.01 m on an occupancy grid. CC and FR disagree in 4 of the 300 cases, indicating sensitivity of the raster reference near narrow or tangential contacts. WCCG matches CC on all 300 scenes with zero false positives and zero false negatives. A query costs 1.07 ms against 1.36 ms (CC) and 243 ms (FR), while additionally supplying the gap graph needed by PIHS.

4) *Comparisons:* Four baselines run on Scenario 1 over 40 randomized trials under the same  $W$ -clearance criterion, object distribution and simulator: **DFS-WCCG** (WCCG-guided search with fixed pushes), **SL-Push** (offline and simulation-validated variants), and **Rec-NAMO** (recursive NAMO) [3]–



Fig. 8. Failure cases by baselines Rec-NAMO, SL-Push and DFS-WCCG.

TABLE III

SCALABILITY AND ROBUSTNESS UNDER OBSTACLE DENSITY, ROBOT-TEAM SIZE, AND EXECUTION-SIDE PHYSICS MISMATCH.

| Case                                                           | Trials | Succ.(%) | $PT$ (s)          | $ET$ (s)          |
|----------------------------------------------------------------|--------|----------|-------------------|-------------------|
| <i>Obstacle density</i> (PushAround, $N=2$ , nominal)          |        |          |                   |                   |
| $M = 15$                                                       | 10     | 100      | $2.5 \pm 0.9$     | $37.1 \pm 21.0$   |
| $M = 30$                                                       | 10     | 100      | $11.7 \pm 3.7$    | $100.9 \pm 34.7$  |
| $M = 45$                                                       | 10     | 100      | $53.4 \pm 24.5$   | $272.7 \pm 97.4$  |
| <i>Robot-team size</i> ( $M=30$ , nominal; 10 trials per cell) |        |          |                   |                   |
| PushAround, $N=3$                                              | 10     | 100      | $30.0 \pm 12.7$   | $97.3 \pm 42.1$   |
| PushAround, $N=4$                                              | 10     | 100      | $29.2 \pm 8.9$    | $110.1 \pm 33.1$  |
| SL-Push, $N=3$                                                 | 10     | 80       | $30.0 \pm 46.3$   | $193.3 \pm 111.3$ |
| SL-Push, $N=4$                                                 | 10     | 80       | $23.4 \pm 18.2$   | $171.7 \pm 61.2$  |
| DFS-WCCG, $N=3$                                                | 10     | 100      | $142.6 \pm 183.1$ | $178.1 \pm 72.1$  |
| DFS-WCCG, $N=4$                                                | 10     | 90       | $101.9 \pm 103.9$ | $175.9 \pm 113.6$ |
| <i>Physics mismatch</i> ( $M=30$ , $N=2$ )                     |        |          |                   |                   |
| Nominal                                                        | 10     | 100      | $16.8 \pm 3.4$    | $104.7 \pm 37.4$  |
| Mass $\times 0.6-1.4$                                          | 40     | 100      | $16.8 \pm 5.4$    | $105.9 \pm 44.6$  |
| Lateral fric. $\times 0.6-1.4$                                 | 40     | 97.5     | $16.5 \pm 5.7$    | $114.0 \pm 48.5$  |
| Spin fric. $\times 0.6-1.4$                                    | 40     | 97.5     | $15.2 \pm 3.6$    | $100.1 \pm 35.7$  |
| Force limit $\times 0.6-1.4$                                   | 40     | 95.0     | $18.9 \pm 10.9$   | $115.9 \pm 58.5$  |
| All perturbed                                                  | 160    | 97.5     | $16.9 \pm 7.0$    | $108.9 \pm 47.4$  |

Each perturbed row scales one parameter; the others stay nominal.

[5], [16]. Table II shows that PushAround reaches the highest success rate (97.5%) and the lowest combined planning-and-execution time (38.9 s). DFS-WCCG needs more than 100 s of planning (exponential branching); SL-Push (offline) plans in 0.06 s but many displacements fail in execution (62.5%); SL-Push (sim) reaches 75.0% at 18.2 s planning; and Rec-NAMO usually fails to build a connected  $W$ -clear corridor in dense clutter, as illustrated in Fig. 8.

5) *Ablation Studies*: Four internal ablations are conducted: **w/o gap-lookahead** (gaps ranked without downstream costs), **w/o mode-cache** (no warm start from validated modes), **w/o quick-pass** (all candidates evaluated by rollout), and **w/o reinsertion** (failed nodes not reconsidered). Table II shows that gap-lookahead and reinsertion mainly improve robustness (success drops by more than 10% when either is removed), whereas mode-cache only lowers success to 92.5%; quick-pass affects efficiency, with simulations rising from 121.5 to 208.2 and planning from 10.3 to 23.6 s.

6) *Scalability and Robustness Analysis*: Generalization is further evaluated in the same  $8 \times 8$  m workspace by scaling the obstacle density  $M$ , the robot-team size  $N$ , and the execution-side physics parameters (Table III and Fig. 10). (I) **Obstacle density**: with two robots and 10 trials per density, PushAround maintains a 100% success rate for all  $M \in [15, 45]$  at 2.5–53.4 s of planning, whereas at  $M=45$  the external methods solve none of the trials and the no-recursion ablation solves 7/10. (II) **Robot-team size**: at  $M=30$ , PushAround solves all ten trials at  $N=2, 3$ , and 4 after the shared mode-generation refinements, which reduce robot-order sensitivity and allow a

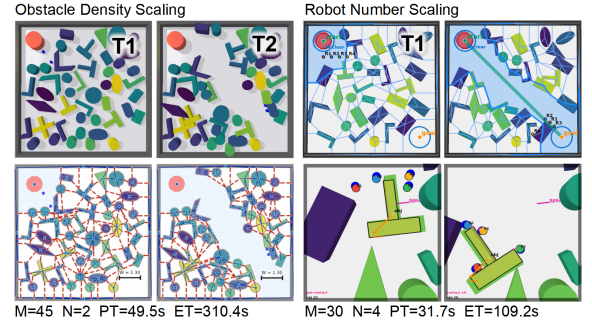


Fig. 9. Scalability of the clearing process. **Left**: obstacle-density scaling ( $M=45$  objects, two robots): snapshots during the clearing (top) and WCCG overlays with the start face in blue (bottom). **Right**: robot-team scaling ( $M=30$  objects, four robots): initial and final states (top) and zoomed views of one push before contact and during execution (bottom).

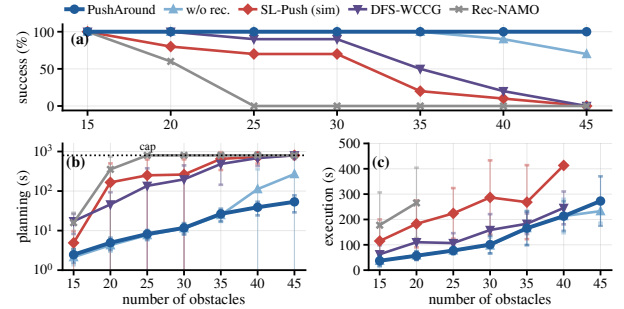


Fig. 10. Obstacle-density scaling ( $8 \times 8$  m, 10 independent seeds per density). (a) Success rate; (b) mean planning time, with unsuccessful trials capped at the 800 s budget (dotted line); (c) mean execution time of the successful trials (Rec-NAMO has none beyond  $M=20$ ). Error bars denote one standard deviation across seeds.

temporary per-task robot release when no full-team assignment exists, with a ranking penalty preferring full-team modes. At  $N=4$ , PushAround uses 29.2 s of planning and 110.1 s of execution, compared with 101.9 s and 175.9 s for DFS-WCCG. (III) **Physics mismatch**: the planner keeps the nominal model and only the execution simulator is perturbed, one parameter at a time. PushAround succeeds in 156 of 160 perturbed trials (97.5%). Under the same  $\pm 40\%$  perturbations, SL-Push (sim) succeeds in 13/40 trials and DFS-WCCG in 23/40 trials; their mean planning times are 31.5 s and 229.1 s, respectively, compared with 18.4 s for PushAround.

## B. Hardware Experiments

1) *System Description*: Experiments use a  $5 \times 6$  m workspace of interlocking  $0.6 \times 0.6$  m foam mats (Fig. 11), 2 UGVs and 6 cardboard-box movable obstacles. Robots and movables each carry three to four motion-capture markers whose global poses are streamed to PyBullet in real time. The two Mecanum UGVs are velocity-controlled, and the optimized contact forces are used only for feasibility evaluation rather than direct motor control. Motion capture provides the current robot and object poses, compared with the successor state predicted by EvalSim: a push triggers replanning only when the position error exceeds 0.3 m or the orientation error exceeds  $\pi/4$ , a loose threshold that tolerates slight over-push or off-axis push. Movables are randomly placed for each trial.

2) *Results*: Two representative real-world runs are shown in Fig. 11. In Scenario 1, PIHS breaks five candidate gaps within 6.5 s using 12 node expansions, 49 visited snapshots and

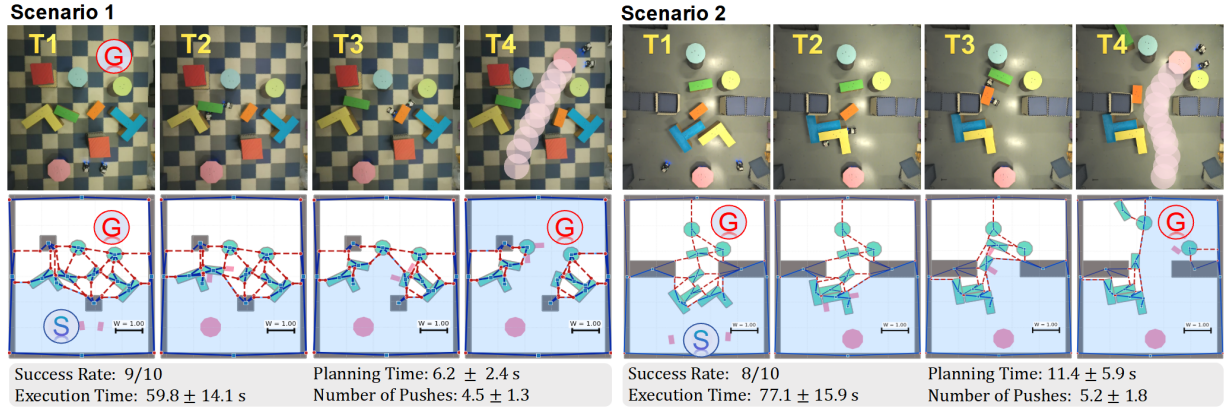


Fig. 11. Real-world pushing experiments with execution-planning alignment (over 20 runs and 2 scenarios). **Top:** snapshots of two robots pushing obstacles in a  $5 \times 6$  m workspace with movable objects and fixed boundaries. **Bottom:** WCCG overlays with the start face in blue.

search depth 6; the four-task schedule first rotates the yellow L-shaped object to open a gap near an immovable obstacle and then separates the green and orange objects until a  $W$ -clear path is established at  $t = 52$  s. In Scenario 2, PushAround succeeds in 8/10 trials with an average of 5.2 pushes and 11.4 s of planning; it first selects a direct push on the blue L-shaped object, which also moves the adjacent T-shaped block and opens the first gap into the narrow passage. This neighboring-object motion is not a separately encoded action but an emergent outcome of the coupled rigid-body rollout in EvalSim, whose snapshot is used to update the WCCG and continue replanning. Execution takes 77.1 s on average, with 15.9 s of deviations from direction-dependent force limits, drifting and Mecanum-wheel slippage, all compensated by online replanning.

## V. CONCLUSION

This work presents PushAround, a multi-robot framework for path clearing in unstructured environments that jointly plans obstacle selection, contact points and pushing forces through a hybrid search, and demonstrates hardware feasibility with state-feedback replanning. Future directions include obstacle-state estimation without external sensing, uncertain masses, and limited robot visibility.

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